

Intermediary Rents in Incentive Contracts

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Abstract

A principal hires an intermediary to incentivize an agent's unobservable effort. Although the intermediary extracts rents, strengthening the agent's outside option allows the principal to reclaim surplus. Consequently, the principal's value and optimal contracts differ markedly from those in standard principal-agent models. Under intermediation, the principal's value strictly increases over a range of the agent's outside option and attains its maximum when that outside option is positive. Furthermore, equilibrium contract payments and effort are non-monotonic and discontinuous in the agent's outside option. My results speak to welfare in labor markets and to the formation of contractual relationships within organizations.

JEL Codes: D23, D82, D86.

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Recent decades have seen a notable rise in contracting with intermediaries. Instead of hiring workers directly, organizations increasingly rely on third parties to manage and incentivize labor. In the private sector, this shift is evident in the expanding use of both outsourcing arrangements and franchising. Firms now outsource not only manufacturing but also information technology and customer support, relying on networks of subcontractors to produce and deliver output.¹ At the same time, many corporations are expanding through franchised stores, where managers independently hire and oversee their own staff.² Intermediated contracting is equally prominent in the public sector, where governments regularly subcontract infrastructure, defense, and other major projects to private firms.

While a vast body of literature already studies contracting in the bilateral firm-worker setting, this paper examines how the presence of an intermediary affects contracting.³ Specifically, I show how the firm’s value, the optimal contracts, and the worker’s effort differ in the intermediated setting compared to the two-party firm-worker framework. To establish a benchmark, consider the case of bilateral contracting. In this setting, both the worker’s compensation and the firm’s profit depend on the worker’s alternative job offers or mandated labor protections—in other words, compensation depends on the worker’s outside options. As his outside options improve, the worker’s bargaining position against the firm strengthens. The wage required to induce his participation increases, and consequently, the firm prefers the worker to have a lower outside option.

This paper’s main point is that when contracting is intermediated, the above results may not hold: workers’ wages can fall as their outside opportunities improve, and firms might strictly prefer workers to have stronger outside options. My findings contrast sharply with those in the bilateral contracting framework and have broad implications for the role of outside options in contracting relationships. In particular, they suggest that—contrary to common perception—outside options are not necessarily a source of conflict between contracting parties. Firms can, in fact, prefer workers to have stronger protections.

My model builds on a canonical principal-agent framework in which a principal values the stochastic output generated by an agent’s unobservable effort. In contrast to most standard models, I assume the principal cannot contract directly with the agent. Instead, the principal must contract with an intermediary, who in turn contracts with the agent. To incentivize the agent’s effort, contracts specify transfers contingent upon the agent’s binary

¹Deloitte’s 2024 Global Outsourcing Survey found that 80% of executives planned to “maintain or increase investment in third-party outsourcing” in 2025.

²The International Franchise Association’s 2025 Franchise Economic Outlook Report predicted that total franchise output would exceed \$936.4 billion in 2025.

³In the moral hazard literature, Holmström (1979) and Innes (1990), among others, characterize optimal contracts in bilateral principal-agent settings. Empirically, several papers study the wage effects of increasing workers’ outside options (Beaudry, Green and Sand 2012; Jäger et al. 2020).

output.⁴ Importantly, the principal cannot condition her contract on the intermediary’s contract with the agent—an assumption that reflects firms’ reluctance to implement such “contracts-on-contracts” in practice. Such arrangements risk exposing firms to liability and are further discouraged by courts’ protection of contract privacy between intermediaries and their employees.⁵

As in the vast literature following Innes (1990), all parties are protected by limited liability. This constraint implies that it is without loss of generality to represent contracts as success-contingent transfers.⁶ Contracting takes place in sequence: the principal first offers a contract to the intermediary, who chooses whether to accept or reject. Upon acceptance, the intermediary offers a contract to the agent, who then chooses whether to accept and, if so, how much effort to exert. If either party rejects their respective contract offers, then all parties receive their outside options. The threat of rejection creates participation constraints for the intermediary and the agent: The value of the principal’s contract must weakly exceed the intermediary’s outside option, with an analogous requirement applying to the intermediary’s contract with the agent.

Compared to standard principal-agent models, the presence of the intermediary enriches the principal’s incentive problem. Instead of simply incentivizing the agent to exert effort, the principal must design a contract that incentivizes the intermediary to induce the desired level of effort from the agent. Notably, because the principal cannot contract directly with the agent, the intermediary can extract a share of the surplus. He does so by offering the agent only a fraction of the principal’s promised transfer, keeping the difference between the principal’s contract and his own as rents. Although this is to the principal’s detriment, the intermediary’s ability to extract rents is not completely unconstrained. Crucially, he must respect the agent’s participation constraint, requiring that the value of his contract weakly exceeds the agent’s outside option.

To illustrate the above discussion, consider a simple example where, upon accepting an offered contract, the agent exerts binary effort, either high or low. High effort incurs a cost of $c = 5$ and yields successful output with certainty; low effort is costless and only yields successful output with probability $1/2$. Moreover, the agent can always take his outside option instead of exerting effort, which yields no output for the principal. To first focus on the intermediary’s rent extraction, suppose that the agent’s outside option is zero. The

⁴For the ease of exposition, my baseline model assumes the agent’s output to be binary. In Section IV, I extend my analysis to the case of continuous output.

⁵Section V discusses this assumption further, and examines how different assumptions on the contracting space affect the results.

⁶This conclusion relies on output being binary. When output is continuous, I show that it is without loss of generality to restrict attention to live-or-die contracts.

agent exerts high effort if his expected utility from doing so exceeds that from exerting low effort, i.e., if he is offered a success-contingent wage w satisfying $w - 5 \geq w/2$. If the principal contracts with the agent directly, she can incentivize high effort with an offer of $w = 10$. However, she concedes rents when contracting is intermediated: upon offering a transfer $t > 0$, the intermediary can obtain profit by offering the agent a smaller wage $w < t$ and obtaining a positive payoff $t - w > 0$ upon the agent's success. Inducing high effort, then, requires the principal to offer a transfer sufficiently high such that the intermediary's expected utility from incentivizing high effort exceeds that from incentivizing low effort, i.e., $t - 10 \geq t/2$. This implies a transfer of $t \geq 20$, which is notably greater than the wage required to induce high effort via direct contracting.

To demonstrate how the principal gains from higher agent outside options, suppose instead that the agent's outside option is 6. Now, the intermediary can only retain the agent by paying a wage of at least 11. At this higher outside option, the principal can (indirectly) induce high effort without conceding rents to the intermediary: when the principal offers a transfer $t = 11$, the intermediary is constrained and can only keep the agent by offering him the entirety of the principal's transfer. The intermediary induces high effort but earns nothing upon the agent's successful output. Thus, this higher outside option allows the principal to induce high effort with a transfer $t = 11$, strictly lower than the transfer $t \geq 20$ required when the agent's outside option is zero.

This example reveals that the agent's outside option constrains the intermediary; in this paper, I investigate both when and how the principal leverages this constraint. My first result, Theorem 1, fully characterizes the principal's value and reveals its dependence on the agent's outside option. The principal's value is initially constant, then strictly increasing, and finally strictly decreasing in the agent's outside option. A key implication of this result is that the principal's value is strictly maximized when the agent's outside option is positive. Then, Theorem 2 characterizes the principal's choice of contract and finds that the equilibrium contracts and effort induced are both non-monotonic and discontinuous in the agent's outside option. When the agent's outside option is low, transfers and effort are constant. Once the agent's outside option becomes sufficiently high, the principal discontinuously decreases her transfer, and the intermediary's wage and the agent's effort decrease accordingly. Beyond this point, the equilibrium transfer, wage, and effort strictly increase with the agent's outside option.

Driving my results is the principal's ability to discipline the intermediary's rent extraction by leveraging the agent's outside option. In particular, over a range of agent outside options, the principal can improve her value by offering the intermediary a *rent-preventing transfer*—the lowest transfer that both prevents the intermediary's rent extraction and allows for the

agent’s retention. To illustrate, consider again the example above. When the agent’s outside option is zero, any positive transfer $t > 0$ would allow the intermediary to offer a strictly lower wage and still attain a positive expected payoff. Thus, the principal can only prevent rent extraction with a transfer of zero, which yields low effort from the agent. In contrast, when the agent’s outside option is 6, the rent-preventing transfer is 11. Upon receiving this transfer from the principal, the intermediary passes the entire transfer to the agent, inducing high effort but extracting no rents.

The agent’s outside option is a key determinant of the optimality of the rent-preventing transfer. When the agent’s outside option is low, the intermediary is constrained only at low transfer levels. Therefore, both the rent-preventing transfer and the effort it induces are low. The principal instead opts for a higher transfer, conceding rents to the intermediary but inducing higher effort. As the agent’s outside option increases, the intermediary becomes constrained at progressively higher transfers. The rent-preventing transfer increases, allowing the principal to (indirectly) incentivize greater effort while still preventing rent extraction. At an intermediate value of the agent’s outside option, a point of discontinuity arises: the principal is indifferent between a high transfer—at which the intermediary extracts rents—and the rent-preventing transfer.⁷ At this point, and for all higher outside options, the principal switches to the rent-preventing transfer, benefiting from the progressively higher effort it induces.

These findings challenge classical predictions of standard principal-agent theory and contribute to ongoing debates surrounding worker benefits and protections. While my analysis abstracts from many aspects of these broad and complex debates, it aims to show how the contracting structure can affect the role of outside options on both firm profits and worker productivity. One central topic of debate surrounds minimum wages. Proponents argue that raising the minimum wage can improve worker well-being, while opponents contend that it is too costly to firms.⁸ Standard principal-agent models contribute to this discussion by predicting that a firm will never benefit from increases in workers’ minimum wages. When contracting is intermediated, I show that higher worker outside options can benefit the upstream firm. Moreover, increasing the worker’s outside option over an intermediate region improves the upstream firm’s payoff even without imposing further losses on the intermediary (Corollary 3). Taken together, my results imply that stronger worker protections can

⁷Because the example above considers only binary effort, it does not exhibit the discontinuity of equilibrium effort in the agent’s outside option. The main model captures this by considering effort chosen from a continuous interval.

⁸Critics also argue that higher minimum wages may reduce employment by increasing labor costs. This paper abstracts from these and related issues, focusing instead on the effect of the contracting structure on incentive provision.

improve welfare without sacrificing firms' profits.

Beyond the implications for firm incentives, this paper's characterization of equilibrium contracts and effort also speaks to questions regarding operational efficiency. Principal-agent models suggest that the principal's transfer, and consequently the worker's effort, increase with the agent's outside option. However, empirical evidence on this relationship is mixed. Some studies find that increasing workers' outside options—for example, through unemployment benefits or outside job opportunities—enhances productivity and increases wages (Coviello, Deserranno and Persico 2022; Hafner and Hedtrich 2024). Conversely, others find that such benefits reduce productivity (Ahammer, Fahn and Stiftinger 2025). My analysis offers a potential explanation for this divergence: when contracting is intermediated, transfers might decrease with the agent's outside option to prevent rent extraction by the intermediary. This, in turn, weakens the worker's incentives, and as a result, effort and productivity decrease.

This paper investigates the role of the agent's outside option in shaping incentives within an intermediated contracting structure. In Section III.B, I show how the mechanisms uncovered can extend beyond this setting. For example, when contracting with multiple intermediaries, outside options arise endogenously: an agent's alternative to one intermediary's contract is the contract offered by another. As Corollary 4 demonstrates, this endogeneity allows the principal to shape, rather than take as given, the agent's alternatives. Specifically, the principal can increase the agent's outside option by contracting with multiple intermediaries. This result provides a rationale for why firms might maintain multiple contracting relationships, and speaks broadly to how contracting networks form.

Related Literature. This paper contributes to the literature on contracting with moral hazard. Traditionally, this literature studies principal-agent models, following the seminal works of Mirrlees (1976), Holmström (1979), Grossman and Hart (1983), and Innes (1990). I build on this framework by studying how equilibrium payoffs and contracts change when contracting occurs via an intermediary. More precisely, my model extends Innes (1990), who studies contracting in a principal-agent setting when the agent is risk-neutral and protected by limited liability.

My paper also relates to work on intermediated contracting in settings without moral hazard. The hierarchical contracting literature typically studies models of adverse selection, where the agent's cost of effort is his private information (Crémer and Riordan 1987; Baron and Besanko 1992; Melumad, Mookherjee and Reichelstein 1995; Laffont and Martimort 1998; Mookherjee and Reichelstein 2001). Two notable exceptions are Baliga and Sjöström (1998) and Macho-Stadler and Pérez-Castrillo (1998), who study intermediation in moral

hazard settings. Broadly, this literature asks when the principal prefers contracting with an intermediary rather than contracting directly with the agent. Absent communication costs, information asymmetry, or the possibility of collusion, the principal generally prefers direct contracting (see Mookherjee, 2012, for a review). My results agree with this finding by showing that in a model with moral hazard, the principal weakly prefers contracting with the agent directly to contracting with the intermediary. That said, my results also emphasize that the extent of the principal’s loss from intermediation depends on the agent’s outside option. In Section II.C, I discuss how my analysis complements existing rationales for intermediated contracting.

Just as this paper studies intermediation in a moral hazard setting, the vertical contracting literature examines intermediation when firms face a pricing problem. An upstream distributor sells a product to a downstream retailer, who then sells the product to consumers (Hart et al. 1990; O’Brien and Shaffer 1992; McAfee and Schwartz 1994; Segal and Whinston 2003).⁹ This setting of successive monopoly gives rise to the classic “double marginalization” problem, in which each firm independently imposes a markup on the product’s price (Spengler 1950). As in this paper, the intermediary’s monopoly power allows him to extract rents. To address this distortion, the vertical contracting literature explores various mechanisms that mitigate double marginalization, such as multi-part tariffs (Oi 1971), maximum retail price maintenance (Overstreet 1983; Shaffer 1991), and increased demand risk (Ghili and Schmitt 2024). These mechanisms play a role similar in spirit to that of the agent’s outside option in my model.

The remainder of this paper is organized as follows. Section I introduces the model. Section II presents the main results, and Section III discusses their application to welfare and to more general contracting environments. Section IV analyzes the case of continuous output. Section V concludes by discussing generalizations of my results.

I. Model

The model extends a standard principal-agent model with moral hazard to include an intermediary. There are three players, all of whom are risk-neutral: a principal (she), an intermediary (he), and an agent (he). The principal benefits from the agent’s productive effort; however, she cannot contract with the agent directly. Instead, the principal must contract with the intermediary, who then contracts with the agent. The intermediary and the agent are protected by limited liability, which prevents contracts from specifying a loss

⁹For a review of the vertical contracting literature, see Tirole (1988) or Katz (1989).

at any output realization. The intermediary’s outside option is zero, and the agent’s outside option is $u \geq 0$. The principal does not have an outside option and is required to hire the agent.¹⁰

The agent produces a stochastic, binary output. He chooses effort $e \in [0, 1)$, where e is the probability of successful output. The cost of effort is $c(e)$, where c is increasing, strictly convex, and thrice continuously differentiable with $c''' \geq 0$.¹¹ To rule out corner solutions, assume no effort is costless and $c'(0) = 0$, $\lim_{e \rightarrow 1} c'(e) = \infty$.

The agent’s effort is private and cannot be contracted upon. I further assume that the principal cannot condition her contract on the intermediary’s contract with the agent. As I discuss in Section V, this assumption reflects the fact that such “contracts-on-contracts” are rarely observed in practice.¹² Instead, both the principal’s and the intermediary’s contracts specify transfers contingent on the agent’s output. Due to limited liability, contracts optimally specify a transfer of zero upon output failure, and it is without loss of generality to represent contracts as success-contingent transfers.

The timing is as follows. First, the principal offers the intermediary a transfer $t \in \mathbb{R}_{\geq 0}$ for the agent’s successful output. The intermediary then chooses to accept or reject the principal’s contract. If he accepts, the intermediary offers the agent a wage $w \in \mathbb{R}_{\geq 0}$ for successful output. The agent then chooses to accept or reject the intermediary’s contract; upon acceptance, he chooses effort e . If either the intermediary or the agent rejects their respective contracts, the game ends, and all players receive their outside options.

The players’ payoffs are as follows. The principal’s value for the agent’s successful output is $y > 0$, and her value for unsuccessful output is zero. The principal’s expected payoff is her value for output net of her transfer to the intermediary in the case of successful output:

$$e(y - t).$$

The intermediary does not value the agent’s output directly, and only values transfers from the principal’s contract. If the intermediary accepts the principal’s contract, his ex-

¹⁰The principal’s outside option does not affect the qualitative insights of the model. Similarly, given limited liability, setting the intermediary’s outside option to zero is a normalization.

¹¹For instance, a quadratic cost function with positive coefficients satisfies all three requirements. The convexity of c ensures the intermediary’s problem is well-behaved, and the convexity of c' does the same for the principal’s problem. Specifically, the convexity of c' guarantees the cost of incentivizing the agent’s effort e is convex in effort, while the convexity of c' guarantees that the principal’s cost of inducing the intermediary to incentivize effort is convex in effort.

¹²An analogous assumption is made in Watson (2024), who observes that courts generally refrain from examining contracts between parties other than those directly involved in the dispute. A similar restriction appears in the double marginalization literature, where analysis often restricts attention to linear contracts. See Riordan (2008) and Loertscher and Marx (2022) for a broader discussion of this assumption in the double marginalization setting.

pected payoff from contracting with the agent is

$$e(t - w).$$

Conditional on accepting the intermediary's contract, the agent's expected payoff from exerting effort e is

$$ew - c(e).$$

I characterize the subgame-perfect Nash equilibria of the above game and analyze equilibrium values, contracts, and effort. For expositional clarity, I assume the principal induces the minimum effort continuation play among her optimal choices. Nevertheless, the conclusions of all my results continue to hold under any other uniform tie-breaking rule.¹³ The players' relevant maximization problems are as follows.

Agent's Problem. Given the intermediary's contract, which specifies a wage w , the agent chooses whether to accept or reject the intermediary's contract. If the agent accepts, he chooses the effort which maximizes his expected utility:

$$\max_e U(e, w) := ew - c(e).$$

If the value of the program above exceeds his outside option, he accepts the intermediary's contract. Otherwise, the agent rejects and receives his outside option u .

Intermediary's Problem. Given the agent's outside option u and the principal's contract specifying transfer t , the intermediary chooses whether to accept or reject the principal's contract. If the intermediary accepts, his value from contracting with the agent is:¹⁴

$$\begin{aligned} (\mathcal{I}_{t,u}) \quad & \max_{w \in \mathbb{R}_{\geq 0}, e \in [0,1]} e(t - w) \\ & \text{s.t.} \quad e \in \arg \max_{e'} U(e', w) \quad (\text{A-IC}) \\ & \quad \quad U(e, w) \geq u. \quad (\text{A-IR}) \end{aligned}$$

The agent's incentive compatibility constraint (A-IC) reflects that the intermediary's choice of effort to induce must maximize the agent's expected utility, while the agent's

¹³Although the principal's SPNE value is unique, her SPNE contract choice is not. Specifically, multiplicity may arise when the agent's outside option is low. In the Online Appendix, I show that all the results continue to hold even when considering the full set of SPNE contracts.

¹⁴The intermediary's problem assumes that when the agent is indifferent between effort levels, ties are broken in the intermediary's favor. However, under the model's assumptions, this indifference never arises.

individual rationality constraint (A-IR) ensures that the intermediary's wage is sufficiently high to guarantee the agent's participation.

Principal's Problem. Given the agent's outside option u , the principal solves:¹⁵

$$\begin{aligned} \max_{t, w \in \mathbb{R}_{\geq 0}, e \in [0, 1)} \quad & e(y - t) \\ \text{s.t.} \quad & (w, e) \text{ solve } (\mathcal{I}_{t, u}) \quad (\text{I-IC}) \\ & e(t - w) \geq 0. \quad (\text{I-IR}) \end{aligned}$$

The intermediary's incentive compatibility constraint (I-IC) is the key distinction between the principal's problem and the intermediary's. Whereas the intermediary can incentivize the agent's effort directly, the principal must incentivize the intermediary to induce the agent's effort. The intermediary's IC constraint reflects this requirement and ensures that given the principal's transfer t , the wage w and effort e are optimal in the intermediary's problem.

The intermediary's individual rationality constraint (I-IR) requires the principal's transfer to be sufficiently high to ensure the intermediary's participation.¹⁶ Call $V(u)$ the value of the above program, i.e., the principal's value from contracting with the intermediary.

I.A. Benchmarks

Before analyzing the principal's value and optimal contract, I discuss two benchmarks that emphasize how the interaction between two frictions—moral hazard and intermediation—drives my results. Each benchmark removes one of these frictions and shows that, in their absence, the characterization of the principal's equilibrium value, contract, and effort induced differ sharply from those in my model. Specifically, the principal never benefits from the agent having a higher outside option, and equilibrium transfers and effort are continuous and weakly increasing in the agent's outside option.

No Moral Hazard Consider a setting without moral hazard, but in which the principal still contracts with the intermediary. Because effort is contractible, the principal and the intermediary condition their transfers directly on the agent's effort. The principal's optimal contract to the intermediary specifies a positive transfer for her desired effort, and zero otherwise. Because the intermediary derives value only from the principal's contract and not

¹⁵As in the intermediary's problem, the intermediary's indifference between effort levels to induce is broken in the principal's favor, though such indifference never arises under the model's assumptions.

¹⁶Note that in SPNE, both the intermediary and the agent accept their respective contracts when indifferent. Otherwise, the principal can increase her transfer slightly to break the indifference.

from the agent's output, he has no incentive to induce effort other than that specified in the principal's contract. The intermediary's contract to the agent similarly specifies a positive transfer for the principal's chosen effort and zero otherwise. Because the agent only receives positive transfers for the principal's preferred effort, this is the effort he exerts.

Because the intermediary is constrained in his choice of effort to induce and the agent is constrained in his choice of effort to exert, the principal can reduce her transfer to prevent either party from extracting rents. As a result, the principal optimally incentivizes the effort that maximizes social surplus, i.e., first-best effort. Because the principal is the sole claimant to this surplus, increasing the agent's outside option can only increase the cost of retaining the agent. The principal's cost of contracting with the agent increases, and as a result, the principal's value decreases in the agent's outside option.

No Intermediary Consider the case in which the principal contracts directly with the agent but cannot observe the agent's effort. In the terminology of the principal-agent literature, this benchmark corresponds to the second-best setting. The principal solves

$$\begin{aligned} & \max_{w \in \mathbb{R}_{\geq 0}, e \in [0,1]} e(y - w) \\ \text{s.t.} \quad & \quad (\text{A-IC}), (\text{A-IR}). \end{aligned}$$

As in the intermediary's problem, the principal is constrained by the agent's IC and IR constraints. Although the principal does not lose rents to the intermediary, the limited liability constraint ensures the agent extracts rents from the principal's contract. Thus, the principal's value is less than in the first-best setting, i.e., when moral hazard is absent and effort is contractible. Let $V_{SB}(u)$ denote the value of the program above.

When the principal is unconstrained by the agent's outside option, she incentivizes the effort that equates her marginal cost of incentivizing effort with her marginal benefit. As the agent's outside option increases, his participation constraint binds, and the principal must increase her transfer to ensure the agent's participation. Although this induces higher effort from the agent, the principal's marginal benefit from higher effort decreases below that of her marginal cost. As a result, the principal's value decreases.

The equilibrium characterization of both benchmarks is summarized as follows.

Lemma 1. *Consider a benchmark setting with either no moral hazard or no intermediary. In both cases, the principal's value (resp. equilibrium transfers and effort) is weakly decreasing (resp. weakly increasing) and continuous in the agent's outside option.*

II. Equilibrium Analysis

Having established the equilibrium values and contracts in the two benchmarks, I next present my main results. I analyze how the agent's outside option shapes payoffs and contracts under intermediation and emphasize departures from both the no moral hazard setting and the standard principal-agent moral hazard framework.

Section II.A begins by characterizing the principal's value (Theorem 1) and the equilibrium contracts and effort (Theorem 2) with respect to the agent's outside option. Section II.B provides the underlying analysis, revealing how the principal's choice of transfer depends on the agent's outside option. Finally, Section II.C compares the outcomes to the second-best benchmark without intermediation: the two coincide when, and only when, the agent's outside option is sufficiently high (Proposition 1). Appendix A contains all proofs for this section.

II.A. Main Results

Recall that in both benchmark settings, the principal's value is always weakly decreasing in the agent's outside option. My main result shows that when contracting is intermediated, this preference reverses: the principal's value strictly increases for a range of agent outside options.

Theorem 1. *The principal's value $V(u)$ is continuous in the agent's outside option u . There exist cutoffs $0 < \underline{u} < u^*$ such that the principal's value $V(u)$ is*

1. *Constant for low values of the agent's outside option $u \in [0, \underline{u}]$,*
2. *Strictly increasing for intermediate values $u \in [\underline{u}, u^*]$, and*
3. *Strictly decreasing for high values $u \geq u^*$.*

Figure 5 illustrates the two key implications of Theorem 1. First, there always exists a region in which the principal's value is strictly increasing in the agent's outside option (i.e., $u \in [\underline{u}, u^*]$). Within this range, the principal strictly prefers the agent's outside option to be higher. Second, the existence of both the constant and the strictly increasing regions implies that the principal's value at u^* strictly exceeds her value at zero. In other words, the principal's value is strictly maximized at a positive value of the agent's outside option.

As discussed in the introduction and as I develop in the sections that follow, this result depends on the ability of both the agent's outside option and the principal's transfer to limit

¹⁶All figures are generated using an example with quadratic costs.

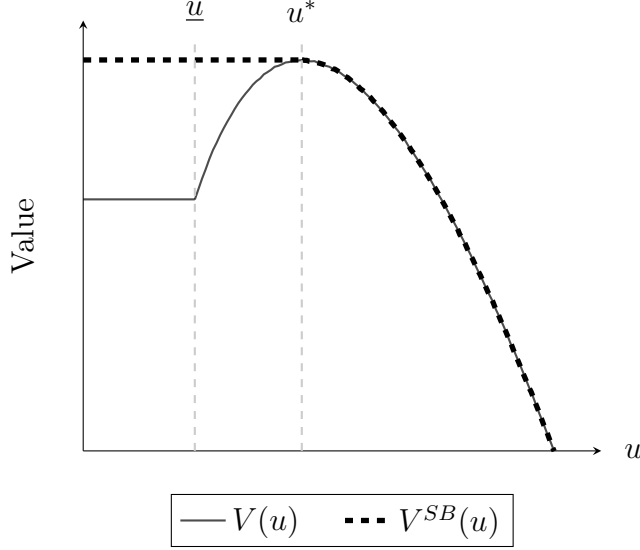


Figure 1: The principal's value with an intermediary (solid) and without (dashed) as functions of the agent's outside option.

the intermediary's rent extraction. When the agent's outside option is sufficiently low, the intermediary can extract rents for a wide range of transfers the principal might offer. As the agent's outside option increases, however, the intermediary must increase his wage in order to retain the agent. As the agent's wage increases, his incentives strengthen, and as a result, effort increases.

While the above logic establishes how higher outside options constrain the intermediary, it remains to clarify the role of the principal's transfer. To do so, I characterize how the optimal transfer, wages, and induced effort vary as the agent's outside option changes. Their characterization contrasts with both benchmarks, in which equilibrium contracts and effort are both monotonic and continuous in the agent's outside option.

Theorem 2. *In equilibrium, the principal's transfer t , the intermediary's wage w , and the agent's effort e are constant for low values of the agent's outside option $u \in [0, \underline{u}]$; decrease discontinuously at $u = \underline{u}$, and are strictly increasing thereafter. For $u \geq \underline{u}$, the principal's transfer and the intermediary's wage are equal, i.e., $t = w$.*

Figure 2 depicts the principal's optimal contract and equilibrium effort. For low values of the agent's outside option, the principal chooses a high transfer that allows the intermediary to extract rents. At an intermediate value, she instead chooses a rent-preventing transfer: a discontinuously lower transfer that constrains the intermediary's rent extraction completely. Theorem 2 emphasizes how, upon receiving this transfer, the intermediary optimally offers a wage equal to the principal's transfer and leaves no rents for himself. The following analysis

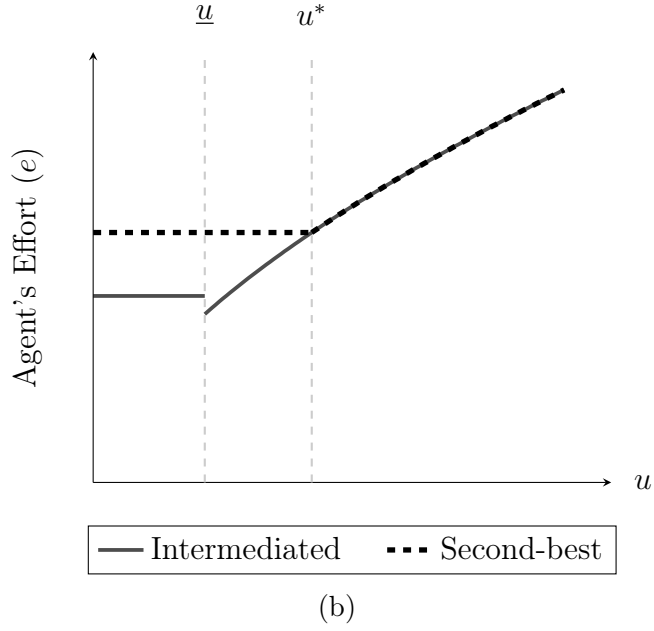
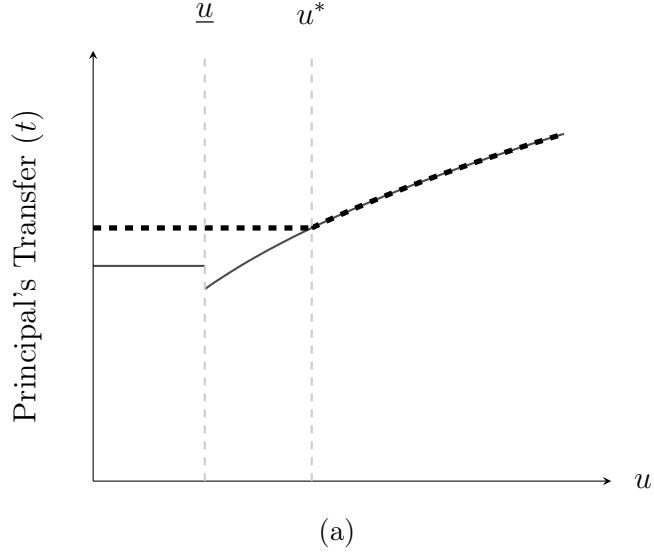


Figure 2: Panel (a) compares the principal's equilibrium transfer with an intermediary (solid) and without (dashed), and Panel (b) presents the corresponding comparison for the agent's equilibrium effort.

investigates both how and when the principal uses this rent-preventing transfer to fully constrain the intermediary.

II.B. Optimal Contracts

The results above are driven by the principal's trade-off between inducing the agent's effort and constraining the intermediary's rent extraction. While higher transfers (indirectly) in-

centivize higher effort, they also allow the intermediary to extract more rents. The agent's outside option plays a crucial role in determining the resolution to this trade-off.

To investigate the principal's choice, I derive her optimal contract and examine how her choice of transfer responds to changes in the agent's outside option. Section II.B.1 begins the analysis by fixing the principal's transfer and characterizing the intermediary's optimal contract. Using this characterization, Section II.B.2 solves for the principal's optimal contract, and shows how the agent's outside option affects her choice of transfer.

II.B.1. Intermediary's Optimal Contract

Within this paper's setting, the intermediary acts as a monopolist: he is the only party able to offer the agent a contract and can therefore extract rents from the principal. However, as I show in this section, his ability to do so is constrained by the agent's outside option.

Suppose the principal offers the intermediary a transfer $t > 0$. Were the intermediary unconstrained by the agent's outside option, he would maximize his value by incentivizing his preferred level of effort $\bar{e}(t)$:

$$\bar{e}(t) := \arg \max_e \left\{ \max_w e(t - w) \text{ s.t. (A-IC)} \right\}.$$

Note that the convexity of cost c and c' ensures that the intermediary's preferred effort $\bar{e}(t)$ is unique and strictly increasing in t . In other words, the intermediary incentivizes higher effort as the principal's transfer increases.

The intermediary, however, cannot always choose to incentivize this effort freely. The agent's IR constraint requires that the agent's expected payoff from the intermediary's contract weakly exceed his outside option. When constrained, the lowest effort the intermediary can induce via a success-contingent wage w is denoted by $\underline{e}(u)$, and defined implicitly by

$$U(\underline{e}(u), w) = u \text{ s.t. (A-IC)}.$$

The agent's IC constraint and the convexity of c ensure that effort $\underline{e}(u)$ is unique and strictly increasing in the agent's outside option u .

The intermediary's choice of effort to incentivize is characterized by the following lemma.

Lemma 2. *If the intermediary accepts the principal's contract offer, he incentivizes effort*

$$e = \begin{cases} \bar{e}(t) & \text{if } t \geq \tau \\ \underline{e}(u) & \text{if } t \leq \tau, \end{cases}$$

by offering the agent a wage $w = c'(e)$. The cutoff transfer τ solves

$$(1) \quad \bar{e}(\tau) = \underline{e}(u).$$

Moreover, the intermediary's rents are positive when $t > 0$ and he incentivizes effort $\bar{e}(t)$.

The intermediary incentivizes either his preferred effort $\bar{e}(t)$, or the effort $\underline{e}(u)$ at which the agent's IR constraint binds. Crucially, $\underline{e}(u)$ is strictly increasing in the agent's outside option. Thus, as the agent's outside option increases, the intermediary optimally induces weakly greater effort. This increase in effort benefits the principal: even without changing her transfer, higher agent outside options force the intermediary to induce higher effort.

II.B.2. Principal's Optimal Contract

The section above illustrated how higher outside options can benefit the principal, even when her contract is fixed. Allowing the principal to choose her contract, then, can further amplify these gains. Following the analysis above, the principal's contracting problem can be captured by a choice between two transfer regimes:

1. In the High Transfers program (HT), the principal offers the intermediary a high transfer $t_{HT} \geq \tau(u)$, where $\tau(u)$ solves equation (1). In this program, the agent's IR constraint is slack, and the intermediary extracts positive rents by incentivizing effort $\bar{e}(t)$.
2. In the Low Transfers program (LT), the principal offers a low transfer $t_{LT} \leq \tau(u)$. In this program, the agent's IR constraint binds, and the intermediary incentivizes effort $\underline{e}(u)$. Within this region, the principal optimally chooses a rent-preventing transfer, which completely prevents the intermediary from extracting rents.

In the following analysis, I characterize how the principal's value and optimal contract in both programs respond to changes in the agent's outside option. Call the principal's value in each program $V_i(u)$ for $i \in \{\text{HT}, \text{LT}\}$. As the two programs encompass all possible transfers the principal can offer, her value equals the maximum over her value in each program:

$$V(u) = \max \{V_{\text{HT}}(u), V_{\text{LT}}(u)\}.$$

Using this characterization, I identify the principal's choice of program in different regions of the agent's outside option.

High Transfers Program. In program (HT), the principal chooses a *high transfer* that (indirectly) induces high effort $\bar{e}(t_{HT})$, but allows the intermediary to extract rents. The principal solves

$$\begin{aligned}
 \text{(HT)} \quad & \max_{t_{HT} \in \mathbb{R}_{\geq 0}} \bar{e}(t_{HT})(y - t_{HT}) \\
 \text{s.t.} \quad & t_{HT} \geq \tau(u). \quad \text{(Prog}_{HT}\text{)}
 \end{aligned}$$

The program constraint (Prog_{HT}) restricts attention to values of the principal's transfer that incentivize the intermediary to induce effort $\bar{e}(t_{HT})$. As Lemma 2 shows, the intermediary extracts strictly positive rents by incentivizing his preferred effort, and thus accepts any contract satisfying (Prog_{HT}).

The principal's program (HT) value $V_{HT}(u)$ is characterized as follows.

Lemma 3. *In program (HT), there exists $\hat{u} > 0$ such that the principal's value $V_{HT}(u)$ (resp. optimal transfer $t_{HT}(u)$) is constant for $u \leq \hat{u}$ and weakly decreasing (resp. increasing) for $u \geq \hat{u}$; and, for $u < \hat{u}$, the optimal transfer strictly exceeds the program cutoff, i.e., $t_{HT}(u) > \tau(u)$. Moreover, given the principal's optimal contract $t_{HT}(u)$, the intermediary's rents are strictly positive.*

In the high transfer regime, increasing the agent's outside option can only increase the intermediary's rents. Thus, the principal's value can never improve as a result of higher outside options. In this respect, the role of the agent's outside option parallels that in the second-best setting: increases in the outside option only reduce the set of feasible transfers, thereby weakly reducing the principal's value.

However, the principal's value in program (HT) is always below that in second-best:

Corollary 1. *For all $u \geq 0$, the principal's value in program (HT) is strictly lower than in second-best, i.e., $V_{HT}(u) < V_{SB}(u)$.*

In program (HT), the intermediary only offers the agent a fraction of the principal's contract. By contrast, when contracting occurs directly, the agent always receives the full value of the principal's contract. Consequently, the principal's cost of incentivizing effort in second-best is always strictly lower than when contracting is intermediated.

Low Transfers Program. In program (LT), the principal offers the intermediary a low transfer that incentivizes $\underline{e}(u)$. The principal's problem is:

$$\begin{aligned}
\text{(LT)} \quad & \max_{t_{LT} \in \mathbb{R}_{\geq 0}} \quad \underline{e}(u)(y - t_{LT}) \\
& \text{s.t.} \quad t_{LT} \leq \tau(u) \quad \text{(Prog}_{LT}\text{)} \\
& \quad \underline{e}(u)(t_{LT} - c'(\underline{e}(u))) \geq 0 \quad \text{(I-IR)}
\end{aligned}$$

The program constraint (Prog_{LT}) restricts attention to values of the principal's transfer that (indirectly) incentivize effort $\underline{e}(u)$, while the intermediary's IR constraint (I-IR) ensures that his utility from accepting the principal's contract is nonnegative. Denoting the principal's value in program (LT) as $V_{LT}(u)$, the next lemma characterizes the optimal transfer and value.

Lemma 4. *In program (LT), the principal's equilibrium transfer and value are*

$$\begin{aligned}
t_{LT}(u) &= c'(\underline{e}(u)), \\
V_{LT}(u) &= \underline{e}(u)(y - c'(\underline{e}(u)))
\end{aligned}$$

For $u > 0$, transfer t_{LT} is strictly below the program cutoff, i.e., $t_{LT}(u) < \tau(u)$; and, the principal's value is strictly increasing for $u \in [0, u^]$, then strictly decreasing for $u \geq u^*$. Moreover, given the principal's optimal contract $t_{LT}(u)$, the intermediary's rents are zero.*

In this program, the principal offers the intermediary a rent-preventing transfer—specifically, she offers the lowest transfer that allows the intermediary to satisfy the agent's IR constraint with equality. The intermediary, bound by the agent's constraint, is then forced to pass the entirety of the principal's transfer directly to the agent (i.e., $t = w$). Although offering this low transfer (indirectly) induces low effort, the principal gains by eliminating the intermediary's rents.

Lemma 4 emphasizes how, by offering this rent-preventing transfer, the principal prevents the intermediary from extracting rents. Nevertheless, the result also shows that the intermediary's presence remains detrimental to the principal relative to second-best. The principal's value in both settings compares as follows:

Corollary 2. *The principal's value in program (LT) compares to that in second-best as follows:*

$$\begin{aligned}
V_{LT}(u) &< V_{SB}(u) \quad \text{for } u \in [0, u^*), \\
V_{LT}(u) &= V_{SB}(u) \quad \text{for } u \geq u^*.
\end{aligned}$$

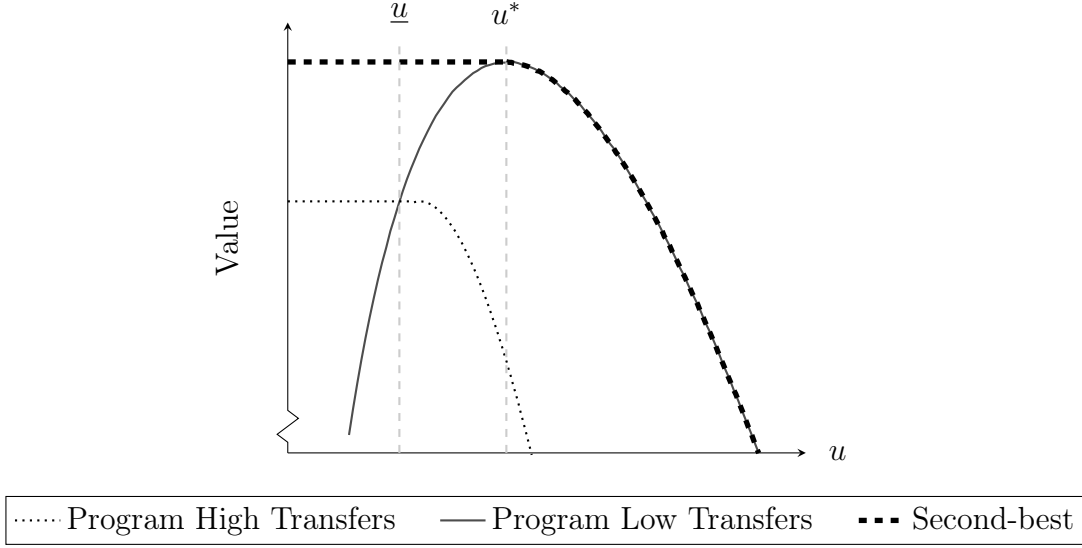


Figure 3: The principal's value from offering a high transfer with intermediary rents (program (HT)), a low transfer that eliminates rents (program (LT)), and contracting directly with the agent (Second-best).

II.B.3. Choice of Regime

The two programs capture the principal's trade-off between a high transfer (program (HT)), and a low transfer (program (LT)). In program (HT), the principal sets a transfer that strictly exceeds the cutoff, $t_{HT}(u) \geq \tau(u)$, whereas in program (LT), the optimal transfer lies below the cutoff, $t_{LT}(u) < \tau(u)$. Thus, the optimal transfers in both programs compare as follows:

$$t_{HT}(u) > t_{LT}(u).$$

While higher transfers are costly for the principal, they can also be beneficial by (indirectly) inducing greater agent effort:

$$e_{HT}(u) \geq e_{LT}(u).$$

However, by allowing the intermediary to implement his preferred effort level, this higher transfer also allows the intermediary to extract rents. In contrast, the principal's rent-preventing transfer in program (LT) prevents the intermediary from extracting rents entirely.

As the next result shows, the resolution of the principal's trade-off between the agent's effort and the intermediary's rents depends on the agent's outside option.

Lemma 5. *There exists $\underline{u} \in (0, u^*)$ such that the principal chooses program (HT) for $u \in$*

$[0, \underline{u})$, and chooses program (LT) for $u \geq \underline{u}$. The principal's value is constant for $u \in [0, \underline{u}]$.

When the agent's outside option is low, the principal chooses program (HT): the loss from lower effort in program (LT) outweighs the loss from the intermediary's rents in program (HT). As the agent's outside option increases, however, effort in program (LT) increases towards the second-best level. This decreases the principal's loss from lower effort in program (LT). At the threshold $\underline{u}$, the gain from removing the intermediary's rents (weakly) outweighs the loss from effort, thereby prompting the principal to switch to program (LT). For outside options beyond this point, effort continues to rise, and the principal's value increases as a result.

To relate the principal's choice of program to the characterization of her value, recall that Theorem 1 identified three distinct regions of the principal's value: constant, strictly increasing, then decreasing. Lemma 5 shows that program (HT) is responsible for the region of constant value, whereas program (LT) explains the region where the principal's value first increases, then decreases.

In a similar spirit, the principal's choice of program explains how equilibrium contracts and effort change in response to increases in the agent's outside option. Because the principal's transfer is always strictly lower in program (LT) than in program (HT), her transfer drops discontinuously when she switches programs (i.e., at $\underline{u}$). This prevents the intermediary from extracting rents, so his wage—and consequently the agent's effort—also decreases.

II.C. The Effect of the Intermediary

Central to my results is the principal's ability to limit the intermediary's rent extraction. Yet, as Theorems 1 and 2 emphasize, the principal's value remains below the second-best level even when the intermediary extracts no rents (i.e., for $u \in [\underline{u}, u^*)$). The next result formalizes how the principal's loss from intermediation depends on the agent's outside option.

Proposition 1. *The principal's value $V(u)$ is strictly less than second-best for $u < u^*$, and equals second-best for $u \geq u^*$.*

Crucially, the principal's loss from intermediation is driven by the intermediary's ability to extract rents when the principal's transfer is sufficiently high—regardless of whether the intermediary does so in equilibrium. In the second-best setting, the principal would prefer to increase her transfer to induce higher effort. However, with an intermediary, she cannot do so without conceding rents. As the agent's outside option increases, this constraint slackens: the principal can remove the intermediary's rents while incentivizing increasingly higher levels of effort. Equilibrium effort increases toward the second-best level, and the principal's value strictly increases as a result.

Although my model abstracts from many aspects of contracting environments, it emphasizes how the agent’s outside option can mitigate inefficiencies introduced by intermediation. My results offer a complementary perspective on what drives different contractual structures, alongside existing rationales such as transaction cost theory (Coase 1937; Williamson 1981, 1985) and property rights theory (Grossman and Hart 1986; Hart and Moore 1990; Hart 1995).¹⁷ Proposition 1 contributes to this discussion by suggesting that—alongside the considerations of these two theories—workers’ outside options may also play a role in firms’ choice of organizational form.

This perspective may help interpret phenomena such as the parallel expansion of the franchise industry and the increase in the minimum wage in the mid-twentieth century (Hunt and Ozanne 1971; Fishback and Seltzer 2021). While each trend is empirically well-documented individually, their connection remains less understood. The framework developed here offers a potential rationale: higher minimum wages strengthen employees’ outside options, as rising market wages and competing opportunities increase workers’ bargaining power. Stronger outside options, in turn, reduce firms’ costs from contracting through intermediaries and consequently make franchising a more lucrative organizational form.

III. Discussion

The results of this paper emphasize that the role of workers’ outside options depends importantly on the contracting structure. To develop how these results can connect to broader discussions of policy design and of how contracting networks form, this section addresses two questions: how rising outside options for workers affect overall welfare, and what drives firms to contract with more than one intermediary.

III.A. Welfare

As discussed in the introduction, debates over worker protections persist. Critics argue that increasing worker protections is detrimental to firms, as it raises labor costs and limits profits. Contrary to this view, Theorem 1 shows that in intermediated contracting structures, upstream firms might benefit when workers have stronger outside options. This result suggests that governments may be able to increase worker protections, such as unemployment benefits or minimum wages, without harming firm profits. However, from the perspective

¹⁷Transaction cost theory predicts that firms outsource to avoid costs associated with search, bargaining, and contract enforcement. Property rights theory takes the approach that contracts are incomplete. Outsourcing is optimal when the intermediary’s ownership of key assets provides stronger incentives for relationship-specific investment.

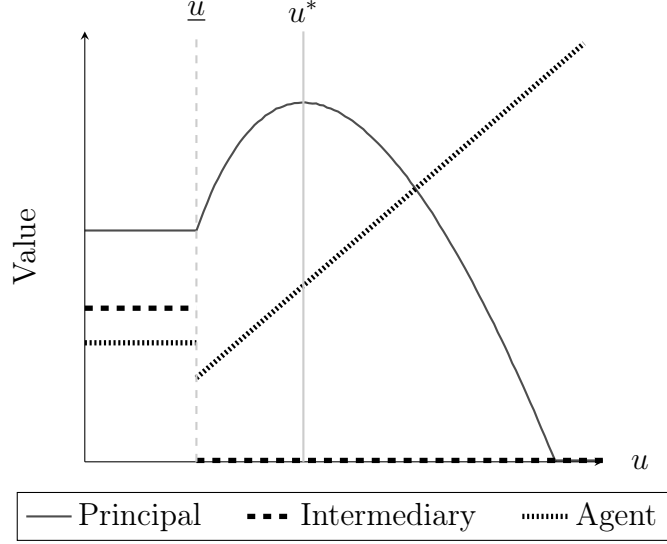


Figure 4: The principal's value (solid), intermediary's value (dashed), and agent's value (dotted) with respect to the agent's outside option.

of a policymaker, a natural question arises: to what extent does the intermediary lose from increasing the workers' outside options? In other words, does increasing the principal's profit come at the expense of the intermediary?

The following result addresses these questions. For an intermediate range of the agent's outside option, not only do the principal and the agent benefit from higher values of the agent's outside option, but the intermediary weakly benefits as well.

Corollary 3. *For $u \in (\underline{u}, u^*]$, increases in the agent's outside option are Pareto-improving.*

Figure 4 illustrates the players' values. In this intermediate region of outside options, increases in the agent's outside option force the intermediary to increase his transfer to the agent. The agent benefits from this higher pay and exerts higher effort. As a result, the principal's value increases. Moreover, recall from Theorem 2 that in this region, the principal optimally constrains the intermediary's rents to zero. Thus, the intermediary is indifferent to higher outside options in this range, as his rents are constrained to zero regardless.

III.B. Endogenizing Outside Options via Competition

Contracting structures often extend well beyond those with one intermediary. In practice, contracting in networked environments is rarely monopolistic: firms routinely contract with multiple intermediaries, who then compete over contracting relationships with downstream firms. Within a supply chain, for instance, a firm may contract with several suppliers that compete over access to the same labor pool. This raises the question of how a firm selects

among potential suppliers and how it balances the resulting contracting relationships. The following extension proposes a rationale for why a principal might prefer contracting with multiple intermediaries rather than maintaining exclusivity with just one.

Consider a contracting structure with two intermediaries. As before, the principal cares about the output of the agent, but cannot contract with him directly. The principal can contract with either or both intermediaries to incentivize the agent's effort, and both intermediaries can contract with the agent. The agent can accept only one contract, so the intermediaries must compete to hire the agent. In this model of intermediary competition, the contract offered by one intermediary serves as the outside option that the agent faces when evaluating the other intermediary's contract offer. The following result builds on the intuition of Theorem 1, and shows that the principal endogenously increases the agent's outside option by offering both intermediaries a nonzero transfer.

Corollary 4. *Suppose the principal can contract with two intermediaries. For $u < u^*$, the principal's contract to both intermediaries specifies positive transfers upon successful output.*

This result demonstrates how the principal can strategically increase the agent's outside option by contracting with multiple intermediaries. By doing so, the principal induces competition between the two intermediaries, thereby reducing their ability to extract rents. While the two-intermediary setup is intentionally simple to illustrate core ideas, the intuition can be generalized to more complex contracting environments. This suggests that competition within a contracting structure may not only reflect market conditions but can also be a deliberate feature of a principal's contracting arrangement.

IV. Continuous Output

While the main model assumes the agent's output to be binary, I show that my results remain robust to relaxing this assumption. In particular, I extend the model to a setting in which the agent's output is continuous. Under assumptions analogous to those made in the binary output case, all the main results extend. All proofs for this section can be found in Appendix B.

The model builds on that of Innes (1990).¹⁸ An agent exerts effort $e \in [0, \bar{e})$ to produce output $y \in [0, \bar{y}]$. The cost of effort is given by $c(e)$, where the properties of c remain the same as in the main model. Output is drawn according to the thrice continuously differentiable conditional probability density function $f(y | e)$, with the corresponding cumulative distribution function $F(y | e)$. I assume that $f(y | e) > 0$ for all $y \in [0, \bar{y}]$ and $e \in [0, \bar{e})$.

¹⁸Note that Innes (1990) assigns the bargaining power to the agent, whereas here it lies with the principal.

The principal's and intermediary's contracts specify transfers contingent on the agent's output and are subject to two-sided limited liability constraints. These constraints ensure that transfers neither exceed the principal's value of the agent's output nor fall below zero. Specifically, the principal's output-contingent transfer that she offers to the intermediary is restricted to $t(y) \in [0, y]$, and similarly, the intermediary's output-contingent wage that she offers to the agent is restricted to $w(y) \in [0, y]$. The contracting structure and the timing of the game remain the same.

Before presenting the results, I introduce two assumptions that are standard in the moral hazard literature. I maintain these assumptions throughout this section.

Monotone Likelihood Ratio Property (MLRP). *For all $y' > y$, $f(y' | e)/f(y | e)$ is strictly increasing in e .*

Convexity of the Distribution Function Condition (CDFC). *For all e , $F(y | e)$ is convex in e .*

MLRP implies that higher output is more informative of higher effort from the agent. CDFC ensures that, given that the agent's wage is monotonically increasing in output, his expected utility is concave in his effort. Together, they are known as the Mirrlees–Rogerson conditions, and are commonly used to solve principal–agent problems via the first-order approach (FOA) (Rogerson 1985; Mirrlees 1999). While their framework assumes a risk-averse agent, an analogous analysis applies when the agent is risk-neutral and protected by limited liability. Building on their results and those of Innes (1990), I show that MLRP and CDFC together ensure that if the principal's contract is monotonically increasing in output, so too is the intermediary's contract. Given CDFC, this monotonicity guarantees that the agent's expected utility is concave in effort, thereby validating the use of the FOA.

Although the monotonicity of the principal's contract is not immediate, I show that restricting attention to monotonically increasing contracts is without loss of optimality both when the principal is unconstrained by the agent's outside option and when she contracts directly with the agent. The next result compares the principal's value at these two points and shows that the principal can, in fact, achieve her second-best value at a positive agent outside option. Recall that u^* is defined in Theorem 1 as the outside option at which the principal's value is maximized.

Proposition 2. *Consider the model of continuous output. The principal's value is strictly maximized at a positive agent outside option. Specifically, $V(u^*) > V(0)$ at $u^* > 0$.*

The intuition parallels that of the binary-output case. At the outside option u^* , the principal reduces her contract value enough to prevent the intermediary from capturing

rents. Because the intermediary is constrained by the agent's outside option, he must pass the full value of the principal's contract directly to the agent. Consequently, the principal attains her second-best value.

MLRP and CDFC alone suffice to show that the principal strictly prefers a positive agent outside option. To extend the remainder of the results to the continuous-output case, I introduce two additional assumptions that—analogueous to the convexity of c and c' in the binary output setting—ensure that the intermediary's problem is well-behaved.¹⁹

Assumption 1. *For $f_e(y | e) > 0$ and $y' > y$, $f_e(y' | e)/f_e(y | e)$ is increasing in e .*

Assumption 2. *For all e , $F_e(y | e) + \bar{y}^{-1}c'(e)$ is convex in e .*

Just as MLRP and CDFC are sufficient for the agent's problem to admit a unique solution, Assumptions 1 and 2 ensure the same for the intermediary's problem. In particular, suppose the principal's contract is monotonically increasing in output. In the principal-agent problem, CDFC ensures that the agent's marginal cost of *exerting* effort is increasing relative to his marginal benefit. Thus, the agent's expected utility is strictly concave with respect to his chosen effort. Playing a similar role, Assumptions 1 and 2 ensure that the intermediary's marginal cost of *incentivizing* effort is increasing relative to his marginal benefit. Then, the intermediary's expected utility is strictly concave in his choice of effort to incentivize. Together, these assumptions are sufficient to ensure that when the principal's contract is monotonically increasing in output, the solution to the intermediary's problem is unique.

Again, the monotonicity of the principal's contract is not immediate. However, I show that for low and intermediate values of the agent's outside option, Assumptions 1 and 2 are sufficient to establish its optimality.

Lemma 6. *Consider the model of continuous output, and suppose Assumptions 1 and 2 hold. For $u \leq u^*$, there exist cutoffs z_i , for $i \in \{P, I\}$, such that the principal's and the intermediary's equilibrium contracts take the following form:*

$$t(y) = \begin{cases} y & \text{if } y \geq z_P, \\ 0 & \text{if } y < z_P, \end{cases} \quad \text{and} \quad w(y) = \begin{cases} y & \text{if } y \geq z_I, \\ 0 & \text{if } y < z_I. \end{cases}$$

¹⁹LiCalzi and Spaeter (2003) discuss two distributions that satisfy MLRP and CDFC, defined for $y \in [0, 1]$:

$$F(y | e) = y + \frac{y - y^2}{e + 1},$$

$$F(y | e) = y \left(\frac{1 - y}{e + 1} + 1 \right).$$

These distributions also satisfy Assumption 1. Assumption 2 is satisfied when c is sufficiently convex relative to the parameters of the distribution: for example, $\bar{y} = 1$, $F(y | e) = y \left(\frac{1 - y}{1 + e} + 1 \right)$, $c(e) = e^4 + e^3$.

The characterization of the equilibrium contracts as “live-or-die” is analogous to the contract characterization in Innes (1990). Above a cutoff, the recipient of the contract receives the full value of the agent’s output; below the cutoff, he receives nothing. The role of the cutoff z_i is analogous to that of the transfer (resp. wage) in the binary-output setting. Specifically, just as increasing the transfer t (resp. w) increases the share of surplus allocated to the intermediary (resp. agent), decreasing the cutoff z_P (resp. z_I) achieves the same effect. The “live-or-die” structure of the principal’s and intermediary’s contracts provides the strongest incentives for the intermediary to incentivize, and for the agent to exert, high effort.

The proof first establishes the validity of the FOA in the intermediated setting, building on the approach of Rogerson (1985).²⁰ I construct an auxiliary program whose value weakly bounds the principal’s intermediated value from above. Then, I show that the principal can attain this upper bound in her original problem. To do so, I first establish that for low and intermediate values of the agent’s outside option (i.e., $u \in [0, u^*]$), the intermediary’s equilibrium contract is monotonically increasing in output. This holds because, under MLRP and CDFC, the contract that incentivizes effort e at the lowest expected cost is a monotonically increasing live-or-die contract. Hence, it is only when the agent’s IR constraint binds that the intermediary might not choose such a contract. In that case, the principal can, and strictly prefers to, offer a live-or-die contract that prevents the intermediary from extracting rents.

Building on this characterization, the auxiliary program maximizes the principal’s expected payoff subject to choosing a contract-effort pair that maximizes the intermediary’s utility over monotonically increasing live-or-die contracts. Given the previous observation,

²⁰In the principal-agent framework, Rogerson establishes the validity of the FOA by first relaxing the principal’s problem—replacing the agent’s IC constraint with his first-order condition—and then showing that the solution to this relaxed problem is feasible in the constrained one. With an intermediary, however, both constructing a relaxation of the principal’s problem and solving a relaxation of the intermediary’s problem are not immediate.

The primary challenge in relaxing the principal’s problem lies in establishing the intermediary’s first-order conditions without first verifying the validity of the FOA for the agent. Moreover, the principal’s problem cannot be relaxed sequentially—by first relaxing the intermediary’s problem, then using his first-order conditions to relax the principal’s problem. This is because, given an arbitrary contract from the principal, the solution to the intermediary’s relaxed problem may not solve his constrained problem. As such, contracts feasible in the principal’s original problem may not be feasible in the sequential relaxation.

This difficulty would resolve if the solution to the intermediary’s relaxed problem were feasible in his constrained problem. However, such feasibility is not immediate in the intermediated setting. Although the intermediary’s problem is structurally similar to the principal’s second-best problem, Rogerson’s approach does not carry over. His proof assumes the principal’s value for the agent’s output to be monotonically increasing in output. Here, by contrast, the intermediary’s value is endogenously determined by the principal’s contract. Thus, without first establishing the monotonicity of the principal’s contract, Rogerson’s approach cannot be applied.

any contract w and effort e that solve the principal's original problem also maximize the intermediary's value over live-or-die contracts and are thus feasible in the auxiliary problem. Under Assumptions 1 and 2, I show that there exists a monotonically increasing live-or-die contract that attains the principal's value from the auxiliary program.

The final step in the proof is to show that the principal's and the intermediary's equilibrium contracts are uniquely live-or-die. For this purpose, I construct a second auxiliary program in which the optimal contracts from the first auxiliary program are feasible. I show that the unique solution to this auxiliary program is a live-or-die contract, thus establishing the uniqueness of the principal's optimal contract for low and intermediate outside options.

Having established the optimality of live-or-die contracts, I now extend the results of my baseline model to the case of continuous output. I show that the principal's value is strictly increasing for intermediate values of the agent's outside option (as in Theorem 1); and that the principal is indifferent between contracting with an intermediary and contracting directly with the agent for high values of the agent's outside option (as in Proposition 1).²¹

Theorem 3. *Consider the model of continuous output. Under Assumptions 1 and 2, the conclusions of Theorem 1 and Proposition 1 extend.*

As in the binary output case, the principal's problem can be represented as a choice between two programs: one in which the intermediary extracts rents, and one in which he does not. The optimality of each program, then, depends on the agent's outside option. For low values of the agent's outside option, the principal allows the intermediary to extract rents in order to achieve higher levels of effort, while for intermediate values of the agent's outside option, she removes the intermediary's rents by decreasing the value of her contract. As the agent's outside option increases, the principal can restrict the intermediary's rent extraction while incentivizing higher levels of effort. It follows that the characterizations of equilibrium contracts and effort mirror those of the binary output case.

Theorem 4. *Consider the model of continuous output. Under Assumptions 1 and 2, the agent's effort e (resp. the principal's cutoff z_P and intermediary's cutoff z_I) are constant for $u \in [0, \underline{u}]$; decrease (resp. increase) discontinuously at $\underline{u}$; and, are strictly increasing (resp. decreasing) for $u \in [\underline{u}, u^*]$.*

By discontinuously increasing her cutoff z_P , the principal prevents rent extraction. Consequently, the intermediary increases his own cutoff z_I , and the agent's effort decreases.

²¹Note that the existence of a contract satisfying both two-sided limited liability and the agent's individual rationality constraint is not guaranteed for all values of the agent's outside option. Nevertheless, nonexistence arises only when the agent's outside option is large enough to render the principal's value negative. Thus, as in the binary-output case, there always exists a range of outside options over which the principal's value is strictly increasing.

V. Concluding Remarks

This paper studies how equilibrium payoffs and contracts depend on the agent’s outside option in intermediated contracting environments. The main results show that in the presence of an intermediary, both contrast sharply with those in standard principal-agent models. First, the principal’s value is increasing for intermediate values of the agent’s outside option; and second, the equilibrium contract and effort exerted are non-monotonic in the agent’s outside option. Together, these results suggest that the role of outside options in influencing incentives and payoffs depends crucially on the contracting structure.

The key mechanism driving my results is the principal’s ability to reduce the intermediary’s rents by leveraging the agent’s outside option. To capture this mechanism, the main model is intentionally simple. Nonetheless, the results are robust to a wide range of settings. The remainder of the conclusion discusses how the results extend beyond the baseline model. The Online Appendix contains formal statements and proofs.

Contracting Space. The analysis focuses on settings in which the principal cannot condition her transfers on the intermediary’s contract with the agent. In other words, the principal cannot write “contracts-on-contracts.” As noted in the introduction and in Section I, this structure is a widely observed feature of contracting relationships. Firms routinely avoid imposing restrictions on subcontractors’ employment contracts in order to limit potential liability for workers, and agreements between intermediaries and agents are typically confidential and protected by nondisclosure agreements or judicial enforcement.

The analysis can extend to environments where the firm has limited discretion over the intermediary’s contract with the agent. Examples of such environments include government contracting or franchising agreements, where principals require intermediaries to provide a base pay to their employees.²² Formally, suppose the principal can choose $\underline{w} \in \mathbb{R}_{\geq 0}$, and can specify transfers contingent on both the agent’s output and whether the intermediary pays the agent at least $\underline{w}$ in each output realization. In such settings, the main results continue to hold. This is because raising $\underline{w}$ constrains the intermediary; however, because the intermediary must now pay $\underline{w}$ even upon output failure, providing incentives becomes more costly. Thus, the principal’s value is maximized by setting $\underline{w} = 0$, and the analysis from the main model applies unchanged.

However, the effects highlighted in this paper arise only when the principal is constrained

²²It is useful to distinguish base pay from the minimum wage discussed in the main text. There, the minimum wage functions as the agent’s outside option, reflecting the minimum utility the agent can obtain at an alternative job. Here, by contrast, base pay is an instrument in the principal’s contract and does not determine the agent’s outside option.

in her ability to write “contracts-on-contracts.” Absent such constraints, the principal can achieve her second-best value by conditioning the transfer w_{SB} on both the agent’s success and the intermediary’s success-contingent wage being w_{SB} . Because the intermediary is constrained by the principal’s contract, he induces second-best effort e_{SB} with wage w_{SB} , and extracts no rents in the process. The analysis then reduces to that of direct contracting.

Intermediary’s Effort. The main model assumes that the principal requires only the productive effort of the agent. Suppose instead that successful output depends on effort from both the intermediary and the agent. The agent exerts effort $e_A \in [0, 1]$ to produce output $q_A \in \{0, 1\}$. The intermediary exerts effort $e_I \in [0, 1]$ to produce output $q_I \in \{0, 1\}$. Let both parties face the same cost of effort $c(e)$, and let their respective probabilities of success be $e_i = \mathbb{P}[q_i = 1 \mid e_i]$ for $i \in \{A, I\}$. The principal’s and the intermediary’s contracts specify transfers contingent on output $q := q_A \times q_I$. The principal offers a contract to the intermediary. If the intermediary accepts, he chooses effort e_I , then offers a contract to the agent. If the agent accepts, he chooses effort e_A . Outputs q_A and q_I are realized, and transfers are made in accordance with the contracts.

In this case, the principal’s value is strictly maximized at a positive value of the agent’s outside option. With intuition similar to that in the main model, increasing the agent’s outside option induces the intermediary to increase his transfer to the agent. Consequently, the agent’s effort increases. Due to the complementarity between the intermediary’s and the agent’s efforts, the intermediary also increases his effort. The principal strictly benefits from increased effort from both parties, and thus strictly prefers the agent to have a positive outside option.

Risk Aversion and Limited Liability. Suppose the agent is risk-averse and protected by limited liability. When the convexity of the agent’s cost increases sufficiently quickly at high effort levels,²³ the main conclusion remains unchanged: the principal’s value is strictly maximized at a positive value of the agent’s outside option. However, whether the result extends without limited liability is not immediate. In the main model, limited liability ensures that increasing the agent’s outside option forces the intermediary to strengthen the agent’s incentives, thereby increasing the agent’s effort. Without limited liability, increasing

²³Formally, I assume the logarithmic derivative of c'' is bounded below by $(1 - e)^{-1}$. This additional convexity assumption is sufficient to ensure that the principal’s expected cost of inducing effort is convex within each of the two contracting regimes the principal might utilize. Specifically, in contrast to the baseline model, the principal offers either (i) a success-contingent transfer, as in the risk-neutral limited liability case, or (ii) both success- and failure-contingent transfers, as in the risk-averse case without limited liability. This assumption ensures the principal’s problem within each regime is well-behaved and is satisfied, for example, by $c(e) = -\ln(1 - e)$.

the agent's outside option may instead induce the intermediary to incentivize lower effort from the agent.

When the agent is risk-averse and not protected by limited liability, the principal's gain from a higher outside option depends on how the intermediary's expected cost of inducing effort, $K(e, u)$, varies with u . If $\frac{\partial^2 K}{\partial e \partial u} < 0$, increases in the agent's outside option reduce the marginal cost of incentivizing higher effort and thus allow the principal to benefit. This can occur, for example, when the agent's utility exhibits constant relative risk aversion (CRRA). Conversely, if $\frac{\partial^2 K}{\partial e \partial u} > 0$, higher outside options raise the marginal cost of incentivizing higher effort and thus harm the principal. This is the case when the agent's utility exhibits constant absolute risk aversion (CARA).

More Intermediaries. Consider a chain of multiple intermediaries. A principal contracts with an upstream intermediary, the upstream intermediary contracts with a downstream intermediary, and the downstream intermediary contracts with the agent. Just as in the binary output case, the principal can leverage the agent's outside option to prevent rent extraction from both the upstream and the downstream intermediary. By offering the upstream intermediary exactly the transfer such that the agent's IR constraint holds with equality, the principal can prevent rent extraction by both intermediaries while also incentivizing positive effort.

A natural next step is to examine even more general contracting networks. While a full analysis is beyond the scope of this paper, a promising direction for future research is to study how outside options shape incentives in richer contracting structures. Such work could shed light on the formation and structure of supply chains and outsourcing arrangements. For instance, how does the ability of a firm to induce competition influence its decision to outsource to many intermediaries rather than a few? Can the rise of unemployment benefits and worker protections help explain the emergence of multi-layered supply chains and hierarchical structures? These questions remain open for future investigation.

Appendix A: Proofs for Section II

Let $(t^*(u), w^*(u), e^*(u))$ denote the minimal effort SPNE outcome at u , and let $(w_{SB}^*(u), e_{SB}^*(u))$ denote the same for the principal's second-best problem.

Proof of Theorem 1. The comparative statics in the principal's value follow from her choice of program. By Lemma 5, the principal chooses program (HT) for $u \in [0, \underline{u})$ and her value is constant; and, for $u \geq \underline{u}$, she chooses program (LT). By Lemma 4, the principal's value strictly increases for $u \in [\underline{u}, u^*]$, then strictly decreases for $u \geq u^*$. $\square$

Proof of Theorem 2. Consider $u \in [0, \underline{u})$. By Lemma 5, program (HT) is optimal. Then, by Lemma 3, the principal's transfer is constant and weakly greater than the cutoff $\tau(u)$, so $\lim_{u \rightarrow \underline{u}^-} t^*(u) = t_{HT}(\underline{u}) \geq \tau(\underline{u})$. Now, for $u \geq \underline{u}$, Lemma 4 shows that the principal's optimal transfer in program (LT) satisfies $t_{LT}(u) = c'(e(u)) < \tau(u)$. The inequalities together imply $\lim_{u \rightarrow \underline{u}^-} t^*(u) > \lim_{u \rightarrow \underline{u}^+} t^*(u)$, establishing the discontinuity in the principal's transfer.

To characterize equilibrium effort, recall that by Lemma 3, (Prog.-HT) is slack for $u < \hat{u}$. Because $\underline{e}(u)$ is strictly increasing in u and Lemma 5 implies $\underline{u} < \hat{u}$, it follows that $\underline{e}(\underline{u}) < \underline{e}(\hat{u}) = \bar{e}(\tau(\hat{u}))$, where the equality follows from the definition of τ . The principal's program choice implies $e^*(u) = \underline{e}(u)$ for $u \geq \underline{u}$ and $e^*(u) = \bar{e}(\tau(\hat{u}))$ for $u < \underline{u}$. Analogously to the characterization of the principal's transfer t , the discontinuity in the agent's effort e follows. The characterization of wage w follows from (A-IC) and the convexity of c . $\square$

Proof of Proposition 1. Recall that Corollaries 1 and 2 compare the principal's value in each program to her value in second-best. Lemma 5 completes the characterization of the principal's value by characterizing her choice of program for each u . $\square$

Before presenting the proofs for Section II.B, I characterize the principal's second-best value and optimal contract. Because this result parallels Lemma 1, their proofs are presented jointly. Define $e^{SB} := e_{SB}^*(0)$, $t^{SB} := t_{SB}^*(0)$, and $u^* := U(e_{SB}, c'(e_{SB}))$.

Claim 1. *The principal's value in second-best $V_{SB}(u)$ (resp. wage w) is constant for $u \leq u^*$, and strictly decreasing (resp. increasing) for $u \geq u^*$. For $u \geq u^*$, (A-IR) binds.*

Proofs of Claim 1 and Lemma 1. The proofs follow from classic arguments used in principal-agent models; see Section 5 in Salanié (2005) for discussion of a similar framework. $\square$

Proof of Lemma 2. Fixing the principal's transfer t , the intermediary's problem can be solved using standard methods in principal-agent models. Characterization of the optimal contract and effort follows from the KKT conditions and the convexity of c and c' .

It remains to show that for $t > 0$, the intermediary's payoff is positive when he incentivizes $\bar{e}(t)$. Because c' is continuous and strictly increasing from $c'(0) = 0$, the Intermediate Value Theorem (IVT) implies $\exists \epsilon > 0$ such that $c'(\epsilon) < t$. Because the intermediary's value weakly exceeds his positive payoff from incentivizing ϵ , his value must be positive. $\square$

Proof of Lemma 3. Implicit differentiation of equation (1) yields $\tau'(u) = \underline{e}'(u)(\bar{e}'(\tau(u)))^{-1}$, where the convexity of c and c' allows for the differentiability of $\underline{e}$ and $\bar{e}$ and implies $\tau'(u) > 0$.

Define the minimum transfer that solves program (HT) as $\underline{t} := \min\{\arg \max_t \bar{e}(t)(y - t)\}$, and define $\hat{u}$ implicitly as $\tau(\hat{u}) = \underline{t}$. Now, consider $u \leq \hat{u}$. Because $\tau(u)$ is strictly increasing in u , $\tau(u) \leq \tau(\hat{u})$ for $u \leq \hat{u}$, with the inequality strict for $u < \hat{u}$. This implies $\underline{t} \geq \tau(u)$, so $\underline{t}$ satisfies (Prog_{HT}). Because $\underline{t}$ is optimal at $u = 0$ and feasible at $\hat{u}$, transfer $\underline{t}$ solves program (HT) and the principal's value in this region is constant at $\bar{e}(\underline{t})(y - \underline{t})$.

That $\frac{\partial V_{HT}}{\partial u} \leq 0$ for $u \geq \hat{u}$ follows from the Envelope Theorem. By Lemma 2, the intermediary extracts positive rents. $\square$

Proof of Corollary 1. By Lemma 2, the intermediary's value in program (HT) is strictly positive, so it must hold that $w = c'(e_{HT}) < t_{HT}$. Thus, the principal can do strictly better in second-best by attaining the same effort e_{HT} with a strictly lower wage of $c'(e_{HT})$. $\square$

Proof of Lemma 4. The KKT stationarity condition yields:

$$-\underline{e}(u) - \lambda + \eta \underline{e}(u) = 0,$$

where λ, η are the Lagrange multipliers for (Prog_{LT}) and (I-IR), respectively. By dual feasibility, the above condition requires $\eta > 0$. This implies (I-IR) binds, so the intermediary's rents are zero and the principal's optimal transfer $t_{LT}(u) = c'(\underline{e}(u))$.

It follows immediately that the principal's value is $V_{LT}(u) = \underline{e}(u)(y - c'(\underline{e}(u)))$, where:

$$\frac{\partial V_{LT}}{\partial u} = \underline{e}'(u) \left[y - \underbrace{(c'(\underline{e}(u)) + \underline{e}(u)c''(\underline{e}(u)))}_{\text{Term (A)}} \right].$$

The convexity of c and c' imply Term (A) is strictly increasing in u . It follows from $\underline{e}'(u) > 0$ that if $\frac{\partial V_{LT}}{\partial u} \big|_u \leq 0$ then $\frac{\partial V_{LT}}{\partial u} \big|_{u'} < 0$ for $u' > u$.

It remains to show that the region of strict increase exists. By the definition of u^* , $V_{LT}(u^*) = V_{SB}(u^*)$. Because $V_{LT}(0) = 0 < V_{SB}(u^*)$, the claim above implies the principal's value must be strictly increasing for $u \in [0, u^*]$. For $u \geq u^*$, Claim 1 shows (A-IR) binds in second-best; this implies that $V_{LT}(u) = V_{SB}(u)$ and both are strictly decreasing in u .

To prove $t_{LT}(u) < \tau(u)$, consider $u > 0$. By Lemma 2, upon receiving transfer $\tau(u)$, the intermediary incentivizes $\bar{e}(\tau(u))$ with wage $w^* = c'(\bar{e}(\tau(u)))$. By Lemma 2, the intermediary's rents are positive, so $\tau(u) > w^*$. From the definition of τ , we also have $\bar{e}(\tau(u)) = \underline{e}(u)$, so $t_{LT}(u) = c'(\underline{e}(u)) = w^*$. The result follows from both relations. $\square$

Proof of Corollary 2. The result follows from Claim 1 and Lemma 4. $\square$

Proof of Lemma 5. To show the principal's switch to program (LT), recall that by Lemmas 3 and 4, $V_{HT}(u)$ is weakly decreasing and $V_{LT}(u)$ is strictly increasing for $u \in [0, u^*]$. Moreover, $V_{LT}(0) = 0$ and because $y > 0$, $V_{HT}(0) > 0$. By Lemma 3, $V_{HT}(u) < V_{SB}(u^*)$ for all u and by Lemma 4, $V_{LT}(u^*) = V_{SB}(u^*)$. Because Lemma 4 and Corollary 2 imply $V(0) < V_{SB}(u^*)$, it follows from the IVT that $\exists \underline{u} < u^*$ such that $V_{LT}(u) \geq V_{HT}(u)$ for $u \geq \underline{u}$.

To show the region of constant value, define $\tilde{u} := \min\{u \mid t_{HT}(u) = \tau(u)\}$ and define $\tilde{t} := t_{HT}(\tilde{u})$. By definition of $\tilde{t}$, V_{HT} is constant and $\tilde{t}$ is optimal for $u \leq \tilde{u}$. Now, recall Lemma 4 shows $t_{LT}(\tilde{u}) < \tau(\tilde{u})$ and that the definition of τ implies $\underline{e}(\tilde{u}) = \bar{e}(\tau(\tilde{u}))$. Thus, $V_{LT}(\tilde{u}) = \underline{e}(\tilde{u})(y - t_{LT}(\tilde{u})) > \bar{e}(\tau(\tilde{u}))(y - \tau(\tilde{u}))$ and the definition of $\tilde{u}$ implies $V_{HT}(\tilde{u}) = \bar{e}(\tau(\tilde{u}))(y - \tau(\tilde{u}))$. It follows that $V_{LT}(\tilde{u}) > V_{HT}(\tilde{u})$. Because $V_{LT}(u)$ is strictly increasing, this implies $\tilde{u} > \underline{u}$. Then, $V_{HT}(u)$ is constant for $u \in [0, \underline{u}]$.

The principal's choice of program (LT) for $u \geq u^*$ follows from Corollaries 1 and 2. $\square$

Proof of Corollary 3. Restrict attention to $u \in [\underline{u}, u^*]$. In this region, Theorem 1 shows that the principal's value is strictly increasing, and Theorem 2 shows the intermediary's transfer equals that of the principal's and is strictly increasing, i.e., $w^*(u) = t^*(u)$. It follows that the intermediary's payoff is zero and the agent's value is strictly increasing. $\square$

Proof of Corollary 4. The principal can offer both intermediaries the transfer $t_{SB} > 0$ upon successful output. Both intermediaries optimally accept the principal's contract, then optimally offer the agent the transfer t_{SB} . The agent randomly chooses one intermediary's contract, and optimally exerts effort e_{SB} . Neither intermediary can deviate to a lower transfer, else the agent rejects the contract and accepts the competing intermediary's. The principal thus achieves his maximum value of $V_{SB}(0)$, which by Proposition 1, strictly exceeds $V(0)$. Note that offering one intermediary a transfer $t < t_{SB}$ allows the other to extract rents; thus, it is strictly suboptimal for the principal. $\square$

Appendix B: Proofs for Section IV

This Appendix discusses the extension to the case of continuous output. The model constrains both the principal and the intermediary by a two-sided limited liability constraint. Define $\mathcal{T} := \{t: [0, \bar{y}] \rightarrow \mathbb{R} \mid t(y) \in [0, y] \forall y \in [0, \bar{y}]\}$ as the set of measurable contracts that satisfy this constraint. The players' problems are written as follows:

Agent's Problem. Upon accepting the intermediary's contract w , the agent chooses effort e to solve $\max_e \left\{ \int_0^{\bar{y}} w(y) f(y \mid e) dy - c(e) \right\}$.

Intermediary's Problem. Upon accepting the principal's contract t , the intermediary solves:

$$\begin{aligned}
 W(t, u) &= \max_{w \in \mathcal{T}, e} \int_0^{\bar{y}} (t(y) - w(y)) f(y \mid e) dy. \\
 (\mathcal{I}_{\text{cont.}}) \quad \text{s.t.} \quad & e \in \arg \max_{e'} \int_0^{\bar{y}} w(y) f(y \mid e') dy - c(e') \quad (\text{A-IC}_{\text{cont.}}) \\
 & \int_0^{\bar{y}} w(y) f(y \mid e) dy - c(e) \geq u \quad (\text{A-IR}_{\text{cont.}})
 \end{aligned}$$

Let $\mathcal{E}_I(t, u)$ denote the set of maximizers for the above program given (t, u) .

Principal's Problem. Given u , the principal's problem is:

$$\begin{aligned}
 V(u) &= \max_{t \in \mathcal{T}, w, e} \int_0^{\bar{y}} (y - t(y)) f(y \mid e) dy \\
 (\mathcal{P}_{\text{cont.}}) \quad \text{s.t.} \quad & (w, e) \in \mathcal{E}_I(t, u) \quad (\text{I-IC}_{\text{cont.}}) \\
 & \int_0^{\bar{y}} (t(y) - w(y)) f(y \mid e) dy \geq 0 \quad (\text{I-IR}_{\text{cont.}})
 \end{aligned}$$

Let $\mathcal{E}_P(u)$ denote the set of maximizers in the principal's problem given u .

Benchmark. As a benchmark, I use a standard principal-agent setup with no intermediary. The principal's value in this second-best case is:

$$\begin{aligned}
 V_{SB}(u) &= \max_{w \in \mathcal{T}, e} \int_0^{\bar{y}} (y - w(y)) f(y \mid e) dy \\
 (\mathcal{SB}_{\text{cont.}}) \quad \text{s.t.} \quad & (\text{A-IC}_{\text{cont.}}), (\text{A-IR}_{\text{cont.}})
 \end{aligned}$$

Let $\mathcal{E}_P^{SB}(u)$ denote the set of maximizers in second-best. To aid the analysis, define $w^{SB} := \min_{(w, e) \in \mathcal{E}_P^{SB}(0)} w$ as the lowest wage which solves $(\mathcal{SB}_{\text{cont.}})$; and, let e^{SB} be the

unique effort, derived from (A-IC_{cont.}), induced by the agent offering w^{SB} .

To further utilize the properties of the principal's value in second-best, define u^* as the maximum outside option at which the principal attains her maximum value:

$$(B.1) \quad u^* := \int_0^{\bar{y}} w^{SB}(y) f(y \mid e^{SB}) dy - c(e^{SB}).$$

Before proceeding to the analysis, define $\bar{u} := \max_{e'} \left\{ \int_0^{\bar{y}} y f(y \mid e') dy - c(e') \right\}$. This is the agent's value if he receives the full value of output. For $u \geq \bar{u}$, the principal and the intermediary cannot satisfy the agent's IR constraint subject to limited liability.

Next, I define a scaled live-or-die contract as the following:

Definition 1. A *scaled live-or-die (l.o.d.) contract* with parameters (α, z) is a contract $g: [0, \bar{y}] \rightarrow \mathbb{R}_{\geq 0}$ that satisfies limited liability and that takes the form:

$$g(y) = \begin{cases} y & \text{if } y \geq z \\ y(1 - \alpha) & \text{if } y < z. \end{cases}$$

A *l.o.d. contract* with cutoff z is a scaled l.o.d. contract with parameter $\alpha = 1$.

For this analysis, I take as given that at $u = 0$, equilibrium effort is strictly positive in both the intermediated and second-best settings. The formal proof is in the Online Appendix.

Proof Overview. First, Section B.1 assumes that the principal's contract is l.o.d., and derives the intermediary's optimal contract. This assumption ensures that the first-order approach can be used when solving the intermediary's contracting problem with the agent.

Section B.2 then solves for the principal's optimal contract. The key step is to verify the validity of the first-order approach (FOA). As discussed in the main text, the method of Rogerson (1985)—relaxing the principal's problem and showing that its solution is feasible in the original problem—does not carry directly to the intermediated setting. To address this, I consider $u \leq u^*$, and show that the intermediary's equilibrium contract must be a l.o.d. contract. Given this observation, I construct an auxiliary program that bounds the principal's value from above. Then, I analyze the auxiliary program by splitting it into two programs, and show that under Assumptions 1 and 2, the principal's optimal contract in the auxiliary program is feasible in her original. To show that the principal's optimal contract is uniquely live-or-die, I use the intermediary's necessary Karush-Khun-Tucker (KKT) conditions to construct another auxiliary program that bounds the principal's value from above. The optimal contract in the auxiliary program is uniquely live-or-die.

Section B.3 concludes by presenting the proofs for the results of Section IV.

B.1. Intermediary's Optimal Contract

In this section, I assume that the principal's contract is a l.o.d. contract, then characterize the intermediary's optimal contract. To solve for the intermediary's optimal contract, I first solve for the lowest expected cost of incentivizing effort e :

$$(Cost) \quad K(e) = \min_{w \in \mathcal{T}} \int_0^{\bar{y}} w(y) f(y | e) dy \quad \text{s.t. (A-IC}_{cont.})$$

To aid the analysis, consider the following relaxation of the above program:

$$(Cost^R) \quad \begin{aligned} K^R(e) &= \min_{w \in \mathcal{T}} \int_0^{\bar{y}} w(y) f(y | e) dy \\ \text{s.t.} \quad &\int_0^{\bar{y}} w(y) f_e(y | e) dy \geq c'(e). \quad (\text{A-IC}_{cont.}^R) \end{aligned}$$

Note that the necessity of the FOC in the agent's problem allows this relaxation.

Before characterizing the solution to (Cost), I define the highest effort which a l.o.d. contract can induce as $e^{UB} := \max \left\{ e \mid \exists z \in [0, \bar{y}] \text{ s.t. } \int_z^{\bar{y}} y f_e(y | e) dy = c'(e) \right\}$.

The following lemma characterizes the solution to (Cost).

Lemma B.1. *A solution to (Cost) exists iff $e \leq e^{UB}$. If $e \leq e^{UB}$, the solution to (Cost) is a l.o.d. contract t_K^* with a unique cutoff z_K^* such that $f_e(z_K^* | e) > 0$ and z_K^* solves*

$$(B.2) \quad \int_{z_K^*}^{\bar{y}} y f_e(y | e) dy = c'(e).$$

Moreover, $K(e) = K^R(e) \forall e$ and $\frac{\partial z_K^*}{\partial e} < 0$.

Proof. To show existence for $e \leq e^{UB}$, define $h(z, e) := \int_z^{\bar{y}} y f_e(y | e) dy - c'(e)$. By MRLP and the definition of e^{UB} , $\exists z^{UB} \in [0, \bar{y}]$ such that $h(z^{UB}, e^{UB}) = 0$. Moreover, by CDFC, h is strictly decreasing in e ; thus, $h(z^{UB}, e) > 0$ for $e \leq e^{UB}$. Note also that $h(\bar{y}, e) < 0$ for $e > 0$. Thus, by the IVT, $\exists z \in [z^{UB}, \bar{y}]$ such that $h(z, e) = 0$ for $e \leq e^{UB}$.

I next show that the solution to the relaxed program (Cost^R) is feasible in, and thus a solution to, the constrained program (Cost). The KKT stationary condition for (Cost^R) is:

$$(B.3) \quad 1 - \lambda_{A-IC} \frac{f_e(y | e)}{f(y | e)} = 0,$$

where $\lambda \geq 0$. Limited liability implies that the optimal contract is a l.o.d. contract, and by MLRP, the cutoff z_K^* is unique. Moreover, dual feasibility implies $\lambda > 0$ and $f_e(z_K^* | e) > 0$. Thus, (A-IC_{cont.}^R) always binds. By CDFC, this contract satisfies (A-IC_{cont.}), and is thus the

constrained problem's unique solution. Note also that the definition of e^{UB} ensures (Cost) does not admit a solution for $e > e^{UB}$.

The comparative static is derived by taking the implicit derivative of equation (B.2):

$$\frac{\partial z_K^*}{\partial e} = \frac{\int_{z_K^*}^{\bar{y}} y f_{ee}(y | e) dy - c''(e)}{z_K^* f_e(z_K^* | e)} < 0.$$

The inequality follows from MLRP, CDFC, and the inequality established above. $\square$

Let $z_K^*(e)$ be the cutoff that incentivizes effort e at the lowest expected cost, as defined in Lemma B.1. The next result is technical and is used to show that when the principal's contract is l.o.d., the intermediary's problem is well-behaved.

Lemma B.2. *Define g as*

$$(B.4) \quad g(e) := \frac{f(z_K^*(e) | e)}{f_e(z_K^*(e) | e)} \left(\int_{z_K^*(e)}^{\bar{y}} y f_{ee}(y | e) dy - c''(e) \right).$$

Under Assumptions 1 and 2, g is strictly decreasing for $e \leq e^{UB}$. Moreover, $g(0) \leq 0$.

Proof. The derivative of g with respect to e is:

$$\begin{aligned} \frac{\partial g}{\partial e} = & \left(\frac{\partial z_K^*}{\partial e} \underbrace{\frac{\partial}{\partial z_K^*} \left[\frac{f(z_K^* | e)}{f_e(z_K^* | e)} \right]}_{\leq 0 \text{ by MLRP}} + \underbrace{\frac{\partial}{\partial e} \left[\frac{f(z_K^* | e)}{f_e(z_K^* | e)} \right]}_{(\text{Term 1}) > 0} \right) \underbrace{\left(\int_{z_K^*}^{\bar{y}} y f_{ee}(y | e) dy - c''(e) \right)}_{\leq 0 \text{ by CDFC}} \\ & + \underbrace{\frac{f(z_K^* | e)}{f_e(z_K^* | e)}}_{\geq 0 \text{ by Lemma B.1}} \left(\underbrace{\left(\int_{z_K^*}^{\bar{y}} y f_{eee}(y | e) dy - c'''(e) \right)}_{(\text{Term 2})} - \frac{\partial z_K^*}{\partial e} \underbrace{\frac{z_K^* f_{ee}(z_K^* | e)}{f_e(z_K^* | e)}}_{(\leq 0 \text{ by Assumption 1 and MLRP})} \right), \end{aligned}$$

where (Term 1) is positive by the following:

$$\frac{\partial}{\partial e} \left[\frac{f(z_K^* | e)}{f_e(z_K^* | e)} \right] = 1 - \frac{f_{ee}(z_K^* | e)}{f_e(z_K^* | e)^2} f(z_K^* | e) > 0,$$

where the inequality follows from Assumption 1 and MLRP.

Consider (Term 2). Integration by parts and the Mean Value Theorem for Integrals yields

$$\begin{aligned} -zF_{eee}(z | e) - \int_z^{\bar{y}} F_{eee}(y | e) dy - c'''(e) &< -zF_{eee}(z | e) - (\bar{y} - z) \min_{z' \in [0, \bar{y}]} F_{eee}(z' | e) dy - c'''(e) \\ &< -\bar{y} \min_{z' \in [0, \bar{y}]} F_{eee}(z' | e) dy - c'''(e). \end{aligned}$$

That (Term 2) is weakly negative then follows from Assumption 2.

Then, the first result follows from $\frac{\partial z_K^*}{\partial e} < 0$, as shown by Lemma B.1. That $g(0) \leq 0$ is implied by the fact that the unique cutoff that satisfies equation (B.2) is $z_K^*(0) = 1$. $\square$

The next lemma shows that to incentivize effort at a cost $K(e)$, it must exceed a cutoff e_C .

Lemma B.3. *Under Assumptions 1 and 2, $\forall u \in [0, u^*] \exists e_C \in [0, e^{SB}]$ that uniquely solves:*

$$(B.5) \quad K(e_C) - c(e_C) = u.$$

For $e \in [e_C, e^{UB}]$, $K(e) - c(e) \geq u$.

Proof. By Lemma B.1, $K(0) - c(0) = 0$. Next, consider the principal's second-best problem at $u = 0$. Because she is unconstrained by (A-IR_{cont.}), her problem reduces to $\max_e \{ \int_0^{\bar{y}} y f(y | e) dy - K(e) \}$. By equation (B.1) and Lemma B.1, $K(e^{SB}) - c(e^{SB}) = u^*$.

It remains to show that $\frac{\partial}{\partial e} [K(e) - c(e)] > 0$ for $e > 0$. Equation (B.3) determines λ_{A-IC} ; applying the Envelope Theorem yields $\frac{\partial}{\partial e} [K(e) - c(e)] = -g(e)$, where g is defined in equation (B.4). By Lemma B.2, $g(e) < 0$ for $e > 0$, so $\frac{\partial}{\partial e} [K(e) - c(e)] > 0$ for $e > 0$. The claim now follows from the IVT. $\square$

Assuming the principal's contract is l.o.d., I next solve the intermediary's problem.

Lemma B.4. *If the principal's contract is a l.o.d. contract t^{LOD} with cutoff z_P , then there exists a scaled l.o.d. contract that solves the intermediary's problem. Moreover,*

1. *If $z_P = z_K^*(e_C)$, where e_C solves equation (B.5), then the intermediary induces effort e_C using a l.o.d. contract w with cutoff $z_K^*(e_C)$. The intermediary's value is zero.*
2. *Consider $u = 0$. Under Assumptions 1 and 2, the intermediary optimally induces effort $e \leq e^{UB}$ by offering the agent a l.o.d. contract w with cutoff $z_K^*(e)$. Effort e is the unique solution to*

$$(B.6) \quad \int_{z_P}^{z_K^*(e)} y f_e(y | e) dy + g(e) = 0.$$

3. If $z_P = 0$, then under Assumptions 1 and 2, any solution to $(\mathcal{I}_{cont.})$ also solves $(\mathcal{SB}_{cont.})$. Moreover, $(A-IR_{cont.})$ binds for $u \geq u^*$; and,

(a) For $u < u^*$, $(A-IR_{cont.})$ is slack and the intermediary induces effort e by offering the agent a l.o.d. contract w with cutoff $z_K^*(e)$. Effort e is the unique solution to

$$\int_0^{\bar{y}} y f_e(y | e) dy - c'(e) + g(e) = 0.$$

(b) $W(t, u)$ is constant for $u \leq u^*$ and is strictly decreasing for $u \geq u^*$.

Proof. The proof closely follows those of Propositions 2-4 and Corollary 3 in Innes (1990). I analyze the following relaxed program $(\tilde{\mathcal{I}}_{cont.})$, then show its solution is feasible in $(\mathcal{I}_{cont.})$:

$$\begin{aligned} W^R(t^{LOD}, u) &= \max_{w \in \mathcal{T}, e} \int_{z_P}^{\bar{y}} y f(y | e) dy - \int_0^{\bar{y}} w(y) f(y | e) dy \\ (\tilde{\mathcal{I}}_{cont.}) \quad \text{s.t.} \quad &\int_0^{\bar{y}} w(y) f_e(y | e) dy \geq c'(e) & (\text{A-IC}_{I-cont.}^R) \\ &\int_0^{\bar{y}} w(y) f(y | e) dy - c(e) \geq u & (\text{A-IR}_{cont.}) \end{aligned}$$

The necessary KKT stationarity conditions are:

$$(B.7) \quad 1 - \lambda_{A-IC} \frac{f_e(y | e)}{f(y | e)} - \lambda_{A-IR} = 0,$$

$$\begin{aligned} (B.8) \quad &\int_{z_P}^{\bar{y}} y f_e(y | e) dy + \lambda_{A-IC} \left(\int_0^{\bar{y}} w(y) f_{ee}(y | e) dy - c''(e) \right) \\ &= \lambda_{A-IR} c'(e) + (1 - \lambda_{A-IR}) \int_0^{\bar{y}} w(y) f_e(y | e) dy \end{aligned}$$

The above conditions yield two candidate solutions:

Candidate 1: $\lambda_{A-IC} > 0$. Limited liability and MLRP imply a l.o.d. contract with cutoff z_I such that $(\text{A-IC}_{I-cont.}^R)$ binds. By CDFC, this solution is feasible in the constrained problem.

To show Lemma B.4.2: If $u = 0$, then $\lambda_{A-IR} = 0$, as $(\text{A-IC}_{cont.})$ does not bind for any $w \in \mathcal{T}$. Then, equation (B.7) implies that $\lambda_{A-IC} > 0$. The result follows from the characterization of the Candidate 1 solution and Lemma B.1. By Lemma B.2, which uses Assumptions 1 and 2, the effort which solves equation (B.8) is unique.

Candidate 2: $\lambda_{A-IC} = 0$. Equation (B.7) implies $\lambda_{A-IR} = 1$; then, equation (B.8) becomes

$$(B.9) \quad \int_{z_P}^{\bar{y}} y f_e(y | e) dy - c'(e) = 0.$$

Denote e^* as the solution to the above. Because $\lambda_{A-IR} = 1$ and (A-IR_{cont.}) binds,

$$(B.10) \quad W^R(t^{LOD}, u) = \int_{z_P}^{\bar{y}} yf(y | e^*) dy - c(e^*) - u.$$

I now claim that if $\lambda_{A-IC} = 0$ at a solution to $(\tilde{\mathcal{I}}_{\text{cont.}})$, then there exists a scaled l.o.d. contract w^{sLOD} with cutoff z_I^* such that (w^{sLOD}, e^*) is feasible, and thus a solution to, $(\mathcal{I}_{\text{cont.}})$. The proof follows analogously to that of Innes (1990). Although Innes (1990) assumes the bargaining power lies with the agent rather than the principal, the structure of the proof remains the same. The claim follows from CDFC.

To show Lemma B.4.1: Suppose $z_P = z_K^*(e_C)$. If the solution is Candidate 2, then the above analysis establishes that there exists a scaled l.o.d. contract that solves $(\mathcal{I}_{\text{cont.}})$. If the solution is Candidate 1, then the optimal contract is a l.o.d. contract. MLRP and CDFC imply the only (scaled) l.o.d. contract that satisfies both (A-IR_{cont.}) and equation (B.6) is a contract with cutoff $z_K^*(e_C)$. Note that this argument did not use Assumption 1 or Assumption 2.

To show Lemma B.4.3: A comparison of the constraints in each program yields $(\mathcal{SB}_{\text{cont.}})$ is equivalent to $(\mathcal{I}_{\text{cont.}})$ when $z_P = 0$. By definition of u^* , (w^{SB}, e^{SB}) as defined in equation (B.1) is feasible for $u \leq u^*$, and therefore solves the intermediary's problem. The intermediary's value is thus constant. By Assumptions 1 and 2, Lemma B.2 implies that the solution when both $\lambda_{A-IC} > 0$ and $\lambda_{A-IR} = 0$ is unique. This implies (A-IR_{cont.}) binds for $u \geq u^*$. By the Envelope Theorem, the intermediary's value is strictly decreasing. $\square$

B.2. Principal's Optimal Contract

First, I establish that in equilibrium, the intermediary's contract is a l.o.d. contract.

Lemma B.5. *Fix $u \leq u^*$ and let $(\tilde{t}, \tilde{w}, \tilde{e}) \in \mathcal{E}_P(u)$. Then, contract $\tilde{w}$ is a l.o.d. contract.*

Proof. I claim that if $\tilde{w}$ is not a l.o.d. contract, then $\tilde{e} \leq e_C$, where e_C is defined in equation (B.5). To prove by contradiction, suppose $\tilde{e} > e_C$. By Lemma B.3, offering a l.o.d. contract with cutoff $z_K^*(\tilde{e})$ satisfies (A-IR_{cont.}), and is thus feasible. By Lemma B.1, a l.o.d. contract strictly minimizes the expected cost of incentivizing $\tilde{e}$. Thus, $\tilde{w}$ cannot be optimal.

Next, I show that the principal strictly prefers incentivizing $e_C \in [0, e^{SB}]$ at cost $K(e_C)$ to incentivizing $e \leq e_C$ at cost $K(e)$. Define $v(e) := \int_0^{\bar{y}} yf(y | e) dy - K(e)$ as her expected utility from incentivizing effort e if she could contract directly with the agent. By the Envelope Theorem and an additional differentiation, $v''(e) = \frac{\partial}{\partial e} \left[\int_0^{\bar{y}} yf_e(y | e) dy - c'(e) \right] + g'(e)$. By CDFC and Lemma B.2, $v''(e) < 0$. Because the principal's second-best value is maximized

at e^{SB} and because the principal is unconstrained at $u = 0$, it holds that $v(e^{SB}) = V_{SB}(0)$. By the concavity of v , $v'(e) > 0$ for $e \leq e^{SB}$.

To conclude the proof, suppose towards a contradiction that $\tilde{w}$ is not a l.o.d. contract. By the first claim, $\tilde{e} \leq e_C$. Then, suppose the principal offers an alternate l.o.d. contract t_K^* with cutoff $z_K^*(e_C)$. By Lemma B.4.1, the intermediary incentivizes effort e_C . By the second claim above and by Lemma B.1, the principal strictly prefers incentivizing e_C at cost $K(e_C)$ to incentivizing $\tilde{e} \leq e_C$ at an expected cost that, by (I-IR_{cont.}), weakly exceeds $K(\tilde{e})$. $\square$

Define the intermediary's problem when maximizing over l.o.d. contracts as:

$$\begin{aligned}
 (\mathcal{I}_{\text{lod}}) \quad & \max_{z_I \in [0, \bar{y}], e} \int_0^{\bar{y}} t(y) f(y | e) dy - \int_{z_I}^{\bar{y}} y f(y | e) dy. \\
 \text{s.t.} \quad & \int_{z_I}^{\bar{y}} y f_e(y | e) dy = c'(e) \quad (\text{A-IC}_{\text{lod}}) \\
 & \int_{z_I}^{\bar{y}} y f(y | e) dy - c(e) \geq u \quad (\text{A-IR}_{\text{lod}})
 \end{aligned}$$

The principal's auxiliary program is defined below, and utilizes the following programs:

1. **Program Slack:** The principal chooses t such that (A-IR_{lod}) is slack in $(\mathcal{I}_{\text{lod}})$. When (A-IR_{lod}) is slack, the KKT conditions imply the following necessary condition:

$$\int_0^{\bar{y}} t(y) f_e(y | e) dy = \int_{z_I}^{\bar{y}} y f_e(y | e) dy - \frac{f(z_I | e)}{f_e(z_I | e)} \left(\int_{z_I}^{\bar{y}} y f_{ee}(y | e) dy - c''(e) \right) \quad (\text{I-IC}_{\text{lod}})$$

Then, the principal solves:

$$\begin{aligned}
 (\text{Slack}) \quad & \max_{t, z_I, e} \int_0^{\bar{y}} (y - t(y)) f(y | e) dy \\
 \text{s.t.} \quad & (z_I, e) \in \text{Slack}(t), \quad (\text{A-IR}_{\text{lod}})
 \end{aligned}$$

where $\text{Slack}(t)$ is the set of (z_I, e) that satisfies the following constraints:

$$\text{Slack}(t) := \{(z_I, e) \mid (\text{I-IC}_{\text{lod}}), (\text{A-IC}_{\text{lod}})\}$$

2. **Program Binds:** The principal chooses t such that (A-IR) binds in $(\mathcal{I}_{\text{lod}})$.

$$\begin{aligned}
& \max_{t, z_I, e} \int_0^{\bar{y}} (y - t(y)) f(y | e) dy \\
\text{(Binds)} \quad & \text{s.t.} \quad (z_I, e) \in \text{Binds}(u) \\
& \int_0^{\bar{y}} t(y) f(y | e) dy - \int_{z_I}^{\bar{y}} y f(y | e) dy \geq 0 \quad (\text{I-IR}_{\text{lod}})
\end{aligned}$$

where $\text{Binds}(u)$ is the set of (z_I, e) that satisfies the following constraints:

$$\text{Binds}(u) = \left\{ (z_I, e) \left| \begin{array}{ll} \int_{z_I}^{\bar{y}} y f_e(y | e) dy = c'(e) & (\text{A-IC}_{\text{lod}}) \\ \int_{z_I}^{\bar{y}} y f(y | e) dy - c(e) = u & (\text{A-IR}_{\text{binds}}) \end{array} \right. \right\}$$

Denote $\tilde{V}_i(u)$ as the principal's value in Program $i \in \{\text{Slack}, \text{Binds}\}$, and $\mathcal{E}_P^i(u)$ as the set of maximizers. The principal's auxiliary problem $(\tilde{P}_{\text{cont.}})$ is defined as follows:

$$(\tilde{P}_{\text{cont.}}) \quad \tilde{V}(u) = \max\{\tilde{V}_S(u), \tilde{V}_B(u)\}.$$

The auxiliary problem is a relaxation of a program in which the principal maximizes her expected payoff *given* the intermediary maximizes over l.o.d. contracts. I now show that for $u \leq u^*$, the principal's value in $(\mathcal{P}_{\text{cont.}})$ is bounded above by her value in $(\tilde{P}_{\text{cont.}})$.

Lemma B.6. *Under Assumptions 1 and 2 and for $u \leq u^*$, if $(t, w, e) \in \mathcal{E}_P(u)$ then w is a l.o.d. contract with cutoff z ; and, (t, z, e) is feasible in either Program Slack or Program Binds.*

Proof. Consider $(\tilde{t}, \tilde{w}, \tilde{e}) \in \mathcal{E}_P(u)$ and define $\mathcal{T}^{LOD}$ as the set of l.o.d. contracts. Restrict attention to $u \leq u^*$. By Lemma B.5, $\tilde{w}$ is a l.o.d. contract with some cutoff $\tilde{z}$, so by CDFC, (A-IC_{cont.}) in $(\mathcal{I}_{\text{cont.}})$ is equivalent to the agent's FOC. Then, because $\mathcal{T}^{LOD} \subseteq \mathcal{T}$, $W(t, u) \geq \tilde{W}(\tilde{t}, u)$. Thus, since $\tilde{w}$ solves $(\mathcal{I}_{\text{cont.}})$ and is feasible in $(\mathcal{I}_{\text{lod}})$, a l.o.d. contract with cutoff $\tilde{z}$ must solve $(\mathcal{I}_{\text{lod}})$. The KKT necessary conditions for $(\mathcal{I}_{\text{lod}})$ yield $(\tilde{z}, \tilde{e}) \in \text{Slack}(\tilde{t}) \cup \text{Binds}(u)$. Moreover, if $(\tilde{t}, \tilde{w}, \tilde{e})$ satisfies (A-IR_{cont.}) in $(\mathcal{I}_{\text{cont.}})$ then it must satisfy (A-IR_{lod}) in $(\mathcal{I}_{\text{lod}})$. Thus, if $(\tilde{t}, \tilde{w}, \tilde{e})$ solves $(\mathcal{P}_{\text{cont.}})$ then $(\tilde{t}, \tilde{z}, \tilde{e})$ is feasible in $(\tilde{P}_{\text{cont.}})$. $\square$

The following results analyze Program Slack and Program Binds.

Lemma B.7. *Under Assumptions 1 and 2, if $(\tilde{t}, \tilde{z}_I, \tilde{e}) \in \mathcal{E}_P^S(u)$ then $\tilde{t}$ is a l.o.d. contract with cutoff $\tilde{z}_P < \tilde{z}_I$ such that $f_e(\tilde{z}_P | \tilde{e}) > 0$. There exists $\hat{u} > 0$ such that $\tilde{V}_S(u)$ is constant for $u \in [0, \hat{u}]$, and weakly decreasing for $u \geq \hat{u}$.*

Proof. By arguments analogous to those in the proof of Lemma B.1, the KKT stationarity conditions, limited liability, and MLRP yield that a l.o.d. contract with cutoff $\tilde{z}_P$ such that $f_e(\tilde{z}_P \mid \tilde{e}) > 0$ uniquely solves (Slack). Moreover, (I-IC_{lod}) requires $\tilde{z}_P < \tilde{z}_I$.

Define $t^* := \max_{(t, z_I, e) \in \mathcal{E}_P^S(0)} t$ and (z_I^*, e^*) such that $(t^*, z_I^*, e^*) \in \mathcal{E}_P^S(0)$. Define $\hat{u}$ as

$$(B.11) \quad \hat{u} := \int_{z_I^*}^{\bar{y}} y f(y \mid e^*) dy - c(e^*).$$

For $u \leq \hat{u}$, (t^*, z_I^*, e^*) is feasible and thus optimal. To complete the proof, the Envelope Theorem yields $\frac{\partial \tilde{V}_S}{\partial u} = -\lambda_{A-IR}$, and by dual feasibility, $\lambda_{A-IR} \geq 0$. $\square$

Lemma B.8. *For $u \in [0, \bar{u}]$, $\tilde{V}_S(u) < V_{SB}(u)$.*

Proof. Let $(\tilde{t}, \tilde{z}_I, \tilde{e}) \in \mathcal{E}_P^S(u)$. Because $(\tilde{z}_I, \tilde{e})$ satisfy (A-IR_{lod}) and (A-IC_{lod}), by CDFC a l.o.d. contract $\tilde{w}$ with cutoff $\tilde{z}_I$ and effort $\tilde{e}$ are feasible in second-best. By Lemma B.7, $\tilde{t}$ is a l.o.d. contract with cutoff $\tilde{z}_P < \tilde{z}_I$. Thus, $V_{SB}(u) \geq \int_0^{\tilde{z}_I} y f(y \mid \tilde{e}) dy > \int_0^{\tilde{z}_P} y f(y \mid \tilde{e}) dy = \tilde{V}_S(u)$. $\square$

Lemma B.9. *Consider $u \in [0, u^*]$. Under Assumptions 1 and 2, if $(\tilde{t}, \tilde{z}_I, \tilde{e}) \in \mathcal{E}_P^B(u)$ then $\tilde{e} = e_C$, where e_C is uniquely defined by equation (B.5) and $e_C < e^{SB}$. Moreover, $\tilde{z}_I = z_K^*(\tilde{e})$. Value $\tilde{V}_B(u)$ and effort $\tilde{e}$ are continuous and strictly increasing in u ; and, $\tilde{V}_B(u^*) = V_{SB}(u^*)$.*

Proof. The KKT stationarity condition for t is $\lambda_{I-IR} = 1$; thus, (I-IR_{lod}) binds and $\tilde{V}_B(u) = \int_0^{\tilde{z}_I} y f(y \mid \tilde{e}) dy$. So, a l.o.d. contract with cutoff z_I satisfies (I-IR_{lod}) and solves (Binds).

To show the region of strict increase, note that by CDFC, (A-IC_{lod}) and (A-IR_{binds}) together yield

$$(B.12) \quad \max_{e'} \left\{ \int_{z_I}^{\bar{y}} y f(y \mid e') dy - c(e') \right\} = u.$$

Denote $\tilde{z}_I(u)$ as the solution to equation (B.12). By the Envelope Theorem, $\tilde{z}_I(u)$ is unique and strictly decreasing in u and by Lemma B.4, $z^{SB} := \tilde{z}_I(u^*) \in [0, \bar{y}]$. That $\tilde{V}_B(u^*) = V_{SB}(u^*)$ follows from CDFC. That $\tilde{z}_I(u)$ is decreasing in u implies $\tilde{z}_I(u) > z^{SB}$ for $u < u^*$. By Lemma B.1, $\forall u$ such that $\tilde{z}_I(u) > z^{SB}$, $\exists e' \in [0, e^{SB})$ such that $\tilde{z}_I(u) = z_K^*(e')$. That $\tilde{e}$ is continuous and strictly increases in u follows from $\tilde{z}_I(u)$ strictly decreasing in u and that, from Lemma B.1, $\frac{\partial z_K^*}{\partial e} < 0$ for $e \leq e^{SB}$.

Recall g as defined in equation (B.4). The KKT stationarity conditions yield

$$\lambda_{A-IR} = \frac{1}{g(e)} \underbrace{\left(g(e) + \int_0^{\bar{y}} y f_e(y \mid e) dy - c'(e) \right)}_{\text{Term (B)}}.$$

By CDFC and Lemma B.2, Term (B) is strictly decreasing in e . By Lemma B.4.3a, e^{SB} is the unique *positive* effort that yields $\lambda_{A-IR} = 0$. By the claim made above, $\tilde{e} < e^{SB}$ for $u < u^*$. Because Term (B) is strictly decreasing in e and $g(e) < 0$ for $e > 0$, it must hold that $\lambda_{A-IR} < 0$ for $u < u^*$. By the Envelope Theorem, $\frac{\partial \tilde{V}_B}{\partial u} = -\lambda_{A-IR} > 0$ in this region. $\square$

The next result compares the principal's value in each program.

Lemma B.10. *Under Assumptions 1 and 2, $\exists \underline{u} \in (0, u^*)$ such that the following hold:*

1. *For $u \in [0, \underline{u})$, $\tilde{V}_S(u) > \tilde{V}_B(u)$ and $\tilde{V}_S(u)$ is constant; and, for $u \geq \underline{u}$, $\tilde{V}_S(u) \leq \tilde{V}_B(u)$ and $\tilde{V}_B(u)$ is strictly increasing in u ;*
2. *For $u \in [0, \underline{u})$, if $(\tilde{t}, \tilde{z}_I, \tilde{e}) \in \mathcal{E}_P^S(0)$ then $(\tilde{t}, \tilde{z}_I, \tilde{e}) \in \mathcal{E}_P^S(u)$;*
3. *For $u \in [0, \underline{u}]$, then for $(t^S, z_I^S, e^S) \in \mathcal{E}_P^S(u)$ and $(t^B, z_I^B, e^B) \in \mathcal{E}_P^B(u)$, $e^S > e^B$.*

Proof. At $u = 0$, because Program Binds yields an effort of zero and, by assumption, the equilibrium effort is greater than zero, $\tilde{V}_S(0) > \tilde{V}_B(0)$. Moreover, by Lemma B.8, $\tilde{V}_B(u^*) = V_{SB}(u^*) > \tilde{V}_S(u) \forall u \geq 0$. By Lemma B.9, $\tilde{V}_B(u)$ is strictly increasing for $u \leq u^*$; that $\exists \underline{u} \in (0, u^*)$ such that $\tilde{V}_B(u) > \tilde{V}_S(u)$ for $u > \underline{u}$ and $\tilde{V}_B(u) \leq \tilde{V}_S(u)$ for $u \leq \underline{u}$ follows.

I next establish the region where $\tilde{V}_S(u)$ and effort e are constant. Take any $(\tilde{t}, \tilde{z}_I, \tilde{e}) \in \mathcal{E}_P^S(0)$ and define $\tilde{u} := \int_{\tilde{z}_I}^{\tilde{y}} y f(y | \tilde{e}) dy - c(\tilde{e})$. By construction, $(\tilde{t}, \tilde{z}_I, \tilde{e})$ solve (Slack) for $u \leq \tilde{u}$, so $\tilde{V}_S(u)$ is constant. If $\tilde{u} > u^*$ the proof concludes; otherwise, Lemma B.9, (A-IR_{binds}), (A-IC_{cont.}) imply $(\tilde{t}', \tilde{z}_I, \tilde{e})$ is feasible in Program Binds, where $\tilde{t}'$ is a l.o.d. contract with cutoff $\tilde{z}_I$. By Lemma B.7, $\tilde{t}$ is a l.o.d. contract with cutoff $\tilde{z}_P < \tilde{z}_I$; thus, $\tilde{V}_B(\tilde{u}) > \tilde{V}_S(\tilde{u})$. By Lemma B.9, $\tilde{V}_B(u)$ strictly increases thereafter. The definition of $\underline{u}$ then implies the region of constant value and that if $(\tilde{t}, \tilde{z}_I, \tilde{e}) \in \mathcal{E}_P^S(0)$ then $(\tilde{t}, \tilde{z}_I, \tilde{e}) \in \mathcal{E}_P^S(u)$. That $e^S > e^B$ for $u \in [0, \underline{u}]$ follows from Lemma B.9. $\square$

Lemma B.11. *If $\tilde{x} := (\tilde{t}, \tilde{w}, \tilde{e}) \in \mathcal{E}_P(0)$ then $\tilde{t}$, $\tilde{w}$ are l.o.d. contracts and $\tilde{x} \in \mathcal{E}_P(u)$ for $u \in [0, \underline{u})$. For $u \in [\underline{u}, u^*]$, if $\tilde{x} \in \mathcal{E}_P(u)$ then $\tilde{t}$, $\tilde{w}$ are l.o.d. contracts with cutoff $z_K^*(\tilde{e})$.*

Proof. By Lemma B.10, the principal chooses Program Slack for $u < \underline{u}$ in $(\tilde{P}_{\text{cont.}})$. Moreover, CDFC and Assumptions 1-2, $\mathcal{E}_P^S(0) = \mathcal{E}_P(0)$. Additionally, by these conditions and Lemma B.10, $(\tilde{t}, \tilde{w}, \tilde{e}) \in \mathcal{E}_P^S(0)$ is feasible, and thus optimal, in $(\mathcal{P}_{\text{cont.}})$ for $u \in [0, \underline{u})$. That $\tilde{t}, \tilde{w}$ are l.o.d. contracts follows from Lemma B.5 and Lemma B.7.

By Lemma B.9 and Lemma B.10, the principal chooses Program Binds for $u \in [\underline{u}, u^*]$. Let $(\tilde{t}, \tilde{z}_I, \tilde{e}) \in \mathcal{E}_P^B(u)$. By Lemma B.4.1, principal can achieve $\tilde{V}_B(u)$ in $(\mathcal{P}_{\text{cont.}})$ by choosing a l.o.d. contract with cutoff $z_K^*(\tilde{e})$. I next show that $\tilde{t}$ is uniquely a l.o.d. contract.

Suppose $(\tilde{t}, \tilde{w}, \tilde{e}) \in \mathcal{E}_P^B(u)$. By Lemma B.6, the value of $(\mathcal{P}_{\text{cont.}})$ is bounded above by the value of $(\tilde{\mathcal{P}}_{\text{cont.}})$, and since $(\tilde{t}, \tilde{w}, \tilde{e})$ is feasible in $(\mathcal{P}_{\text{cont.}})$, it solves $(\mathcal{P}_{\text{cont.}})$. By Lemma B.5, $\tilde{w}$ is a l.o.d. contract and therefore solves $(\mathcal{I}_{\text{lod}})$. The following KKT condition from $(\mathcal{I}_{\text{lod}})$ must be satisfied:

$$\int_0^{\bar{y}} \tilde{t}(y) f_e(y | e) - c'(e) \geq \lambda_{\text{A-IC}_{\text{lod}}} \left[c''(e) - \int_0^{\bar{y}} y f_{ee}(y | e) dy \right],$$

and because $\tilde{w}$ is a l.o.d. contract, it must hold that $\lambda_{\text{A-IC}_{\text{lod}}} \geq 0$ and by CDFC the RHS is positive. Thus, a necessary condition for the principal's contract is:

$$\int_0^{\bar{y}} \tilde{t}(y) f_e(y | \tilde{e}) dy - c'(\tilde{e}) \geq 0. \quad (\text{I-IC}^R)$$

Now, consider the following auxiliary program. For $e \in [0, e^{SB})$, define

$$\tilde{V}(e) := \max_{t \in \mathcal{T}} \int_0^{\bar{y}} (y - t(y)) f(y | e) dy \quad \text{s.t. } (\text{I-IC}^R).$$

Because $\tilde{t}$ is feasible in the above program, $\tilde{V}(\tilde{e}) \geq V(u)$. By analysis analogous to that in the proof of Lemma B.1, the KKT conditions, limited liability, and MLRP ensure that the unique solution to the above program is a l.o.d. contract t^* with cutoff $z_K^*(\tilde{e})$. Moreover, $(t^*, \tilde{w}, \tilde{e})$ is feasible in $(\mathcal{P}_{\text{cont.}})$, and thus must be the unique solution to this program. $\square$

Finally, I establish the solution to the principal's problem for $u \geq u^*$.

Lemma B.12. *For $u \geq u^*$, $V(u) = V_{SB}(u)$ and $V(u)$ is strictly decreasing in u .*

Proof. By (I-IR_{cont.}), (A-IR_{cont.}), and (A-IC_{cont.}), $V_{SB}(u) \geq V(u) \forall u \geq 0$. By Lemma B.4 there exists a scaled l.o.d. contract t and effort e that solves $(\mathcal{SB}_{\text{cont.}})$. By MLRP, CDFC, and Assumptions 1-2, (t, t, e) is feasible in $(\mathcal{P}_{\text{cont.}})$; thus, $V_{SB}(u) \leq V(u)$. By the two inequalities, $V_{SB}(u) = V(u)$. By Lemma B.4.3, $V_{SB}(u)$ strictly decreases for $u \geq u^*$. $\square$

B.3. Proof of Main Results

Proof of Proposition 2. At u^* , the principal can offer w^{SB} to the intermediary, and by Lemma B.4.1 and B.4.3, the intermediary must offer w^{SB} to the agent. By CDFC, the resulting effort is e^{SB} . Because (w^{SB}, e^{SB}) satisfies (A-IR_{cont.}) in second-best, it also satisfies (A-IR_{cont.}) in the intermediated setting. Thus, the principal's value is $V(u^*) = V_{SB}(u^*)$.

At $u = 0$, (I-IR_{cont.}) and that equilibrium effort is positive imply $V(0) < V_{SB}(0)$. By the definition of u^* , $V_{SB}(u^*) = V_{SB}(0)$. The result follows from the two relations. $\square$

Proof of Lemma 6. Follows from Lemma B.11. $\square$

Proof of Theorem 3. The characterization of the principal's value for $u \in [0, u^*]$ follows from Lemma B.11 and Lemma B.9, and for $u \geq u^*$ from Lemma B.12. The comparison to second-best follows from the same results. $\square$

Proof of Theorem 4. The discontinuity in effort and cutoffs follows from Lemma B.11 and Lemma B.10. Their continuity in the region $u \in [\underline{u}, u^*]$ follows from Lemma B.9. $\square$

References

- Ahammer, Alexander, Matthias Fahn, and Flora Stiftinger.** 2025. “Outside Options and Worker Motivation.” *Working Paper*.
- Aneja, Karan, Juan Coronado, and Mike Stoler.** 2024. “Global Outsourcing Survey 2024.” Deloitte.
- Baliga, Sandeep, and Tomas Sjöström.** 1998. “Decentralization and Collusion.” *Journal of Economic Theory*, 83(2): 196–232.
- Baron, David P., and David Besanko.** 1992. “Information, Control, and Organizational Structure.” *Journal of Economics & Management Strategy*, 1(2): 237–275.
- Beaudry, Paul, David Green, and Benjamin Sand.** 2012. “Does Industrial Composition Matter for Wages? A Test of Search and Bargaining Theory.” *Econometrica*, 80(3): 1063–1104.
- Coase, R. H.** 1937. “The Nature of the Firm.” *Economica*, 4(16): 386–405.
- Coviello, Decio, Erika Deserranno, and Nicola Persico.** 2022. “Minimum Wage and Individual Worker Productivity: Evidence from a Large US Retailer.” *Journal of Political Economy*, 130(9): 2315–2360.
- Crémer, Jacques, and Michael H. Riordan.** 1987. “On Governing Multilateral Transactions with Bilateral Contracts.” *The RAND Journal of Economics*, 18(3): 436–451.
- Fishback, Price V., and Andrew J. Seltzer.** 2021. “The Rise of American Minimum Wages, 1912–1968.” *Journal of Economic Perspectives*, 35(1): 73–96.
- Ghili, Soheil, and Matt Schmitt.** 2024. “Risk Aversion and Double Marginalization.” *Journal of Industrial Economics*, 72: 762–806.
- Grossman, Sanford J., and Oliver D. Hart.** 1983. “An Analysis of the Principal-Agent Problem.” *Econometrica*, 51(1): 7–45.
- Grossman, Sanford J., and Oliver D. Hart.** 1986. “The Costs and Benefits of Ownership: A Theory of Vertical and Lateral Integration.” *Journal of Political Economy*, 94(4): 691–719.

- Hafner, Flavio, and Christoph Hedtrich.** 2024. “The Equilibrium Effects of Workers’ Outside Employment Options: Evidence from a Labour Market Integration.” *Working Paper*.
- Hart, Oliver, and John Moore.** 1990. “Property Rights and the Nature of the Firm.” *The Journal of Political Economy*, 98(6): 1119–1158.
- Hart, Oliver, Jean Tirole, Dennis W. Carlton, and Oliver E. Williamson.** 1990. “Vertical Integration and Market Foreclosure.” *Brookings Papers on Economic Activity. Microeconomics*, 1990: 205.
- Hart, Stuart L.** 1995. “A Natural-Resource-Based View of the Firm.” *The Academy of Management Review*, 20(4): 986–1014.
- Holmström, Bengt.** 1979. “Moral Hazard and Observability.” *The Bell Journal of Economics*, 10(1): 74–91.
- Hunt, Shelby D., and Urban B. Ozanne.** 1971. “The Economic Effects of Franchising.” In *The Economic Effects of Franchising*. U.S. Government Printing Office.
- Innes, Robert D.** 1990. “Limited Liability and Incentive Contracting with Ex-Ante Action Choices.” *Journal of Economic Theory*, 52(1).
- Jäger, Simon, Benjamin Schoefer, Samuel Young, and Josef Zweimüller.** 2020. “Wages and the Value of Nonemployment*.” *The Quarterly Journal of Economics*, 135(4): 1905–1963.
- Katz, Michael L.** 1989. “Chapter 11 Vertical Contractual Relations.” In *Handbook of Industrial Organization*. Vol. 1, , ed. Richard Schmalensee and Robert D. Willig, 655–721. Amsterdam:Elsevier.
- Laffont, Jean-Jacques, and David Martimort.** 1998. “Collusion and Delegation.” *The RAND Journal of Economics*, 29(2): 280–305.
- LiCalzi, Marco, and Sandrine Spaeter.** 2003. “Distributions for the First-Order Approach to Principal-Agent Problems.” *Economic Theory*, 21: 167–173.
- Loertscher, Simon, and Leslie M. Marx.** 2022. “Incomplete Information Bargaining with Applications to Mergers, Investment, and Vertical Integration.” *American Economic Review*, 112(2): 616–649.

- Macho-Stadler, Inés, and J. David Pérez-Castrillo.** 1998. “Centralized and Decentralized Contracts in a Moral Hazard Environment.” *The Journal of Industrial Economics*, 46(4): 489–510.
- McAfee, R. Preston, and Marius Schwartz.** 1994. “Opportunism in Multilateral Vertical Contracting: Nondiscrimination, Exclusivity, and Uniformity.” *The American Economic Review*, 84(1): 210–230.
- Melumad, Nahum, Dilip Mookherjee, and Stefan Reichelstein.** 1995. “Hierarchical Decentralization of Incentive Contracts.” *The RAND Journal of Economics*, 26(4): 654–672.
- Mirrlees, J. A.** 1999. “The Theory of Moral Hazard and Unobservable Behaviour: Part I.” *The Review of Economic Studies*, 66(1): 3–21.
- Mirrlees, James A.** 1976. “The Optimal Structure of Incentives and Authority within an Organization.” *The Bell Journal of Economics*, 7(1): 105–131.
- Mookherjee, Dilip.** 2012. “Incentives in Hierarchies.” In *The Handbook of Organizational Economics*. 764–798. Princeton University Press.
- Mookherjee, Dilip, and Stefan Reichelstein.** 2001. “Incentives and Coordination in Hierarchies.” *The B.E. Journal of Theoretical Economics*, 1(1): 20011005.
- O’Brien, Daniel P., and Greg Shaffer.** 1992. “Vertical Control with Bilateral Contracts.” *The RAND Journal of Economics*, 23(3): 299–308.
- Oi, Walter Y.** 1971. “A Disneyland Dilemma: Two-Part Tariffs for a Mickey Mouse Monopoly.” *Quarterly Journal of Economics*, 85(1): 77–96.
- Overstreet, Thomas R.** 1983. “Resale Price Maintenance: Economic Theories and Empirical Evidence.” Federal Trade Commission, Bureau of Economics Staff Report.
- Riordan, Michael H.** 2008. “Competitive Effects of Vertical Integration.” In *Handbook of Antitrust Economics*. MIT Press.
- Rogers, Ashley, and Jin Qi.** 2025. “Franchising Economic Outlook.” International Franchising Association.
- Rogerson, William P.** 1985. “The First-Order Approach to Principal-Agent Problems.” *Econometrica*, 53(6): 1357–1367.

- Salanié, Bernard.** 2005. *The Economics of Contracts: A Primer*. Vol. 1, The MIT Press.
- Segal, Ilya, and Michael D. Whinston.** 2003. “Robust Predictions for Bilateral Contracting with Externalities.” *Econometrica*, 71(3): 757–791.
- Shaffer, Greg.** 1991. “Capturing Strategic Rent: Full-Line Forcing, Brand Discounts, Aggregate Rebates, and Maximum Resale Price Maintenance.” *The Journal of Industrial Economics*, 39(5): 557–575.
- Spengler, Joseph J.** 1950. “Vertical Integration and Antitrust Policy.” *Journal of Political Economy*, 58(4): 347–352.
- Tirole, Jean.** 1988. *The Theory of Industrial Organization*. The MIT Press.
- Watson, Joel.** 2024. “Contractual Chains.” *Econometrica*, 92(5): 1735–1774.
- Williamson, Oliver E.** 1981. “The Economics of Organization: The Transaction Cost Approach.” *American Journal of Sociology*, 87(3): 548–577.
- Williamson, Oliver E.** 1985. “The Economic Institutions of Capitalism.” *New York: Free Press*.

Online Appendix for “Intermediary Rents in Incentive Contracts”

by Michelle Hyun-Kim

The appendix in the main text presents the proofs of the main results. In this Online Appendix, Section A discusses the tie-breaking assumption of choosing the minimal effort equilibrium and shows how Theorem 2 generalizes. Section B provides statements and proofs for extensions discussed in the main text’s conclusion. Section C provides supplementary proofs for the case of continuous output.

A. Equilibrium Contracts

The main text assumes tie-breaking in favor of the minimal effort equilibrium. This section shows that when we consider the set of SPNE outcomes, the main conclusions of Theorem 2 continue to hold. Denote $\mathcal{E} : [0, \infty) \rightarrow \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \times [0, 1)$ as the correspondence which maps each u to the set of (t, w, e) that maximize the principal’s problem.

Definition 2. Consider the correspondence $\hat{\mathcal{E}} : [0, \infty) \rightarrow \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \times [0, 1]$. $\hat{\mathcal{E}}$ is **constant** on the interval $[a, b]$ if, for all $u_0, u_1 \in [a, b]$, $\mathcal{E}(u_0) = \mathcal{E}(u_1)$.

Recall $\underline{u}$ is defined in Theorem 1 as the outside option at which the principal’s value begins to increase. Moreover, recall $t_{LT}(u)$ is defined in Lemma 4 as the principal’s optimal transfer in program (LT). The following result shows that the insights of Theorem 2 carry without a tie-breaking rule:

Proposition A.1. The set of SPNE outcomes $\mathcal{E}(u)$ is compact-valued and constant at $\mathcal{E}(0)$ for $u \in [0, \underline{u}]$; discontinuous at $\underline{u}$; and is single-valued for $u > \underline{u}$, where $\mathcal{E}(u) = \{(t_{LT}(u), t_{LT}(u), \underline{e}(u))\}$. Moreover, the following relations hold:

$$(A.1) \quad \min_{(t,w,e) \in \mathcal{E}(0)} t > t_{LT}(\underline{u}), \quad \min_{(t,w,e) \in \mathcal{E}(0)} w > t_{LT}(\underline{u}), \quad \min_{(t,w,e) \in \mathcal{E}(0)} e > \underline{e}(\underline{u}).$$

Proof. To begin, note that the principal’s choice of program depends only on the principal’s value, and the principal’s SPNE payoff is unique. Moreover, the solution to program (LT) is unique. Thus, Lemmas 3-5 hold without loss. Note also that the Maximum Theorem implies the set of SPNE outcomes in both program (HT) and program (LT) are upper hemicontinuous in u and compact-valued.

By Lemma 5, program (HT) is optimal for $u \in [0, \underline{u}]$. Restrict attention to this region. Define $\tilde{t} := \min_{(t,w,e) \in \mathcal{E}(0)} t$, $\tilde{e} := \bar{e}(\tilde{t})$, and $\tilde{w} := c'(\tilde{e})$. By the characterization of program (HT), $\tilde{x} := (\tilde{t}, \tilde{w}, \tilde{e}) \in \mathcal{E}(0)$ and $\tilde{w} = \min_{(t,w,e) \in \mathcal{E}(0)} w$, $\tilde{e} = \min_{(t,w,e) \in \mathcal{E}(0)} e$.

Define $\tilde{u} := \tilde{w}\tilde{e} - c(\tilde{e})$ as the lowest value of the agent's outside option at which, upon receiving transfer $\tilde{t}$, (A-IR) holds with equality in the intermediary's problem. Because $\tilde{x}$ maximizes the principal's value at $u = 0$, $\tilde{x}$ is feasible at $\tilde{u}$, and because $\tilde{u}$ only decreases the set of feasible $\tilde{x}$ compared to $u = 0$, it follows that $\tilde{x} \in \mathcal{E}(\tilde{u})$. To show $\mathcal{E}$ is constant for $u < \tilde{u}$, note that since $\tilde{t} = \tau(\tilde{u})$, then $\tilde{t}$ satisfies (Prog_{HT}). It follows that $t > \tilde{t}$ satisfies (Prog_{HT}). Thus, all $t \in \mathcal{E}(0)$ are feasible at $\tilde{u}$. It follows that $\mathcal{E}(0) \subseteq \mathcal{E}(u)$ for $u < \tilde{u}$. Moreover, since $\tau(u)$ is weakly increasing in u it follows that $\mathcal{E}(u) \subseteq \mathcal{E}(0)$. Thus, $\mathcal{E}(0) = \mathcal{E}(u)$ for $u \in [0, \tilde{u}]$.

To show the region $u \in [0, \underline{u})$ is constant, notice that by the definition of $\tilde{u}$, transfer $\tilde{t}$ is feasible in program (LT). However, it is not optimal: at $\tilde{u}$, the principal can increase her payoff by offering the intermediary transfer $t < \tilde{t}$ and incentivizing $\tilde{e}$. It follows that $\tilde{e}(y - t) > \tilde{e}(y - \tilde{t})$ for $t < \tilde{t}$, and since $\tilde{t} = \tau(\tilde{u})$, it follows that $V_{\text{LT}}(\tilde{u}) > V_{\text{HT}}(\tilde{u})$. Thus, program (LT) is optimal at $\tilde{u}$ and by Lemma 5, $\underline{u} < \tilde{u}$.

Consider the region $u > \underline{u}$. By the characterization of the principal's choice of program in Lemma 5 and the characterization of program (LT) in Lemma 4, $\mathcal{E}(u) = \{(t_{\text{LT}}(u), t_{\text{LT}}(u), \underline{e}(u))\}$.

The proof for equation (A.1) follows analogously to that of Theorem 2 in the main text.

It remains to show that $\mathcal{E}(u)$ violates lower hemicontinuity at $\underline{u}$. Define $x^{\text{LT}} := (t_{\text{LT}}(\underline{u}), t_{\text{LT}}(\underline{u}), \underline{e}(\underline{u}))$. Moreover, define

$$\epsilon' := \min \left\{ \min_{(t,w,e) \in \mathcal{E}(0)} t - t_{\text{LT}}(\underline{u}), \min_{(t,w,e) \in \mathcal{E}(0)} w - w_{\text{LT}}(\underline{u}), \min_{(t,w,e) \in \mathcal{E}(0)} e - e_{\text{LT}}(\underline{u}) \right\}.$$

By equation (A.1), $\epsilon' > 0$.

At $\underline{u}$, Lemmas 3 and 5 show $\mathcal{E}(\underline{u}) = \mathcal{E}(0) \cup \{x^{\text{LT}}\}$. To show failure of lower hemicontinuity, take any $x^0 \in \mathcal{E}(0)$. Define the open set $X := \{x \mid |x - x^0| < \epsilon'\}$. By construction of ϵ' and by equation (A.1), $\mathcal{E}(0) \cap X \neq \emptyset$ and $\{x^{\text{LT}}\} \cap X = \emptyset$. Now, consider the open set $B_\epsilon(\underline{u}) = \{u \mid |u - \underline{u}| < \epsilon\}$, for any $\epsilon > 0$. Take $u + \frac{\epsilon}{2} \in B_\epsilon(\underline{u})$, and note that the characterization of $\mathcal{E}$ above implies $\mathcal{E}(\underline{u} + \frac{\epsilon}{2}) \cap X = \{x^{\text{LT}}\} \cap X = \emptyset$. Thus, lower hemicontinuity is violated. $\square$

B. Extensions

This section discusses the extensions from the concluding remarks in the main text. Section V discusses alternate contracting spaces. Section B.2 explores the case in which the intermediary can exert effort. Section B.3 considers the case of multiple intermediaries. Section B.4 presents results for the case in which the intermediary is risk-averse.

B.1. Contracting Space

Following the discussion in the main text, suppose the principal can choose $\underline{w} \in \mathbb{R}_{\geq 0}$, and can specify transfers contingent on both the agent's output and whether the intermediary pays the agent at least $\underline{w}$ in each output realization.

Proposition B.1. *Suppose the principal can specify transfers contingent on the intermediary's wage in both output realizations exceeding base pay $\underline{w}$. The conclusions of Theorems 1 and 2 continue to hold.*

Proof. It is without loss of generality that the principal sets her transfer to zero when the intermediary's wage is less than $\underline{w}$, regardless of output realization. Thus, the principal's contract can be represented by $(t_1, t_0, \underline{w})$, where t_1 is the transfer contingent upon both the agent's successful output and $w_1, w_0 \geq \underline{w}$, and t_0 is defined analogously for the agent's unsuccessful output.

Upon receiving $(t_1, t_0, \underline{w})$, the intermediary chooses a contract specifying wages (w_1, w_0) upon the agent's success and failure, respectively. To satisfy the agent's incentive compatibility, the intermediary must choose (w_1, w_0, e) to satisfy

$$w_1 - w_0 = c'(e).$$

Using this expression, the intermediary's expected payoff can be rewritten as

$$(B.1) \quad et_1 + (1 - e)t_0 - ec'(e) - w_0.$$

For any (w_1, w_0, e) that satisfies the agent's IC and IR constraints and the principal's base pay restriction, the intermediary can do strictly better by setting wages $\tilde{w}_0 = \underline{w}$ and $\tilde{w}_1 = w_1 + \frac{1-e}{e}(w_0 - \underline{w})$. Such wages have the same expected cost to the intermediary, and thus continue to respect the agent's IR conditions. Moreover, because $\tilde{w}_1 \geq w_1$ and $\tilde{w}_0 \leq w_0$, the agent exerts weakly higher effort under this contract.

Thus, the intermediary's problem reduces to choosing effort e that maximizes equation (B.1), holding w_0 fixed. Because $\underline{w}$ enters the intermediary's payoff linearly, the solution to the maximization is independent of $\underline{w}$. Thus, the principal's value is weakly maximized by setting $\underline{w} = 0$, and the analysis from the main model applies. $\square$

B.2. Intermediary's Effort

Consider the case in which the intermediary can exert effort. The intermediary exerts effort e_I to produce output $q_I \in \{0, 1\}$. The agent exerts effort e_A to produce output $q_A \in \{0, 1\}$.

For both the intermediary and the agent, the cost of effort function is given by $c(e)$, and the probability of successful output is $\mathbb{P}[q_i = 1 \mid e_i] = e_i$. Denote the joint output between the intermediary and the agent $q := q_I \times q_A$. The timing of the game is as follows. The principal offers a transfer $t \in \mathbb{R}_{\geq 0}$ to the intermediary contingent on joint project success. The intermediary accepts or rejects the contract. Then, the intermediary chooses effort e_I and offers a wage $w \in \mathbb{R}_{\geq 0}$ to the agent contingent on joint project success. If the agent accepts the contract, the agent chooses his effort level e_A . Outputs q_A and q_I are realized, and transfers are made according to the contracts.

Fixing the intermediary's contract w and his effort choice e_I , the agent solves

$$\max_{e_A} e_A e_I w - c(e_A).$$

The agent's expected utility is concave in effort, so his choice of effort solves his FOC:

$$(B.2) \quad w = \frac{c'(e_A)}{e_I}.$$

Fix the principal's transfer t . The intermediary solves

$$\begin{aligned} \max_{w, e_I, e_A} \quad & e_A e_I (t - w) - c(e_I) \\ (\mathcal{I}_{\text{effort}}) \quad \text{s.t.} \quad & w = \frac{c'(e_A)}{e_I} \quad (\text{A-IC}) \\ & e_I e_A w - c(e_A) \geq u. \quad (\text{A-IR}) \end{aligned}$$

The principal's problem is:

$$\begin{aligned} \max_{t, w, e_I, e_A} \quad & e_A e_I (y - t) \\ & (w, e_I, e_A) \text{ solves } (\mathcal{I}_{\text{effort}}) \text{ given } (t, u) \quad (\text{I-IC}) \\ & e_A e_I (t - w) - c(e_I) \geq 0. \quad (\text{I-IR}) \end{aligned}$$

Proposition B.2. *When the intermediary exerts effort, the principal strictly prefers a positive value of the agent's outside option, i.e. there exists $u' > 0$ such that $V(u') > V(0)$.*

Proof. Let (t^0, w^0, e_I^0, e_A^0) solve the principal's problem at $u = 0$. Define $\tilde{u} := e_A^0 c'(e_A^0) - c(e_A^0)$. Given t^0 , consider the intermediary's problem at $\tilde{u}_\epsilon := \tilde{u} + \epsilon$ where $\epsilon > 0$.

Given (I-IC) and (A-IC) in $(\mathcal{I}_{\text{effort}})$, (A-IR) can be rewritten as

$$e_A c'(e_A) - c(e_A) \geq u.$$

Note that by the concavity of f and convexity of c , the LHS of the above is strictly increasing in e_A . Thus, at $u > \tilde{u}$, (A-IR) requires that the intermediary incentivize effort $e_A > e_A^0$.

At $\tilde{u}_\epsilon$, the principal can offer transfer t^0 . By the above argument, the intermediary incentivizes effort $e_A^* > e_A^0$. The KKT conditions for the intermediary's problem yield that the intermediary's choice of effort e_I^* solves:

$$c'(e_I^*) = e_A^* t^0.$$

Because $e_A^* t^0 > e_A^0 t^0$, it follows that $e_I^* > e_I^0$. Thus, the principal can achieve a strictly higher probability of successful output than at $u = 0$, i.e. $e_A^* e_I^* > e_A^0 e_I^0$. Because the intermediary's value is positive at $u = 0$, there exists $\epsilon > 0$ such that for t^0 , (I-IR) is satisfied at t^0 . Thus, $\exists \epsilon > 0$ such that the principal strictly prefers the agent's outside option at $\tilde{u}_\epsilon$ to zero. $\square$

B.3. More Intermediaries

Suppose there are N intermediaries. The principal offers a contract to intermediary I_1 , who then offers a contract to intermediary I_2 , and so on, until intermediary I_{N-1} offers a contract to intermediary I_N , who then offers a contract to the agent.

Proposition B.3. *When there are N intermediaries, the principal strictly prefers a positive value of the agent's outside option.*

Proof. Consider $u = 0$. Because $y > 0$, the principal incentivizes positive effort $e^0 > 0$ and attains positive utility. Denote the set of equilibrium contracts as $(t, w_1, w_2, \dots, w_N)$, where w_i is the contract intermediary i writes to intermediary $i + 1$ for $i < N$ and w_N is the contract written to the agent. Since $e^0 > 0$, it must hold that $w_N > 0$. Moreover, it must also hold that $t > w_N$; otherwise, each intermediary i would offer a contract $w_i < t$ to the next intermediary. Then, intermediary I_N can do better by reducing his contract to $w_N - \epsilon > 0$ for some $\epsilon > 0$, which still satisfies the agent's IR constraint and secures positive expected profits for the intermediary. Define $\tilde{u} := e^0 w_N - c(e^0)$.

At $\tilde{u}$, the principal can offer the contract $\tilde{t} = w_N$. Intermediary I_i must offer $\tilde{w}_i = w_N$ to intermediary I_{i+1} , else the contract will not satisfy the agent's IR constraint. Upon receiving contract w_N , the agent exerts effort e^0 . Because the principal incentivizes the same effort at a strictly lower transfer, $V(\tilde{u}) > V(0)$. $\square$

B.4. Risk Aversion

Consider the case where the agent is risk-averse and the intermediary and principal are risk-neutral. Assume the agent's utility U is twice continuously differentiable, and assume

$U' > 0, U'' < 0, U(0) = 0$. I additionally make the following assumption, which ensures that the minimum cost K of incentivizing effort is convex:

Assumption B.1. *The logarithmic derivative of c'' exceeds $(1-e)^{-1}$, i.e., $\frac{d}{de} [\ln c''(e)] \geq \frac{1}{1-e}$.*

When e is very low, the convexity of cost K is guaranteed by the concavity of U and the convexity of c and c' . This assumption ensures that the costs of incentivizing intermediate and high levels of effort are also convex. A cost function that satisfies this condition, for example, is $c(e) = -\ln(1-e)$.

I first explore the case where the agent is protected by limited liability, and show that the main conclusions continue to hold. Then, I show that without limited liability, whether the conclusions continue to hold depends on the agent's risk aversion.

Before proceeding, consider the intermediary's lowest expected cost of incentivizing effort e when *not* restricted by limited liability:

$$\begin{aligned}
 K(e, u) = \min_{\bar{w}, \underline{w}} \quad & e\bar{w} + (1-e)\underline{w} \\
 (\text{Cost}_{\text{RA}}) \quad & \text{s.t.} \quad c'(e) = U(\bar{w}) - U(\underline{w}) \quad (\text{A-IC}) \\
 & eU(\bar{w}) + (1-e)U(\underline{w}) - c(e) \geq u \quad (\text{A-IR})
 \end{aligned}$$

Note that the concavity of c implies that the agent's FOC is a sufficient condition for (A-IC). The following result characterizes the solution to the above program.

Lemma B.1. *Suppose $U: \mathbb{R} \rightarrow \mathbb{R}$ is such that $U' > 0$ and $U'' < 0$. The unique solution to $(\text{Cost}_{\text{RA}})$ is $(\bar{w}, \underline{w})$ that solves*

$$(B.3) \quad U(\bar{w}) = u + c(e) + (1-e)c'(e),$$

$$(B.4) \quad U(\underline{w}) = u + c(e) - ec'(e).$$

The second and cross-partial derivatives of K , respectively, are given by:

$$(B.5) \quad \frac{\partial^2 K}{\partial e \partial u} = \frac{1}{U'(\bar{w})} - \frac{1}{U'(\underline{w})} + e(1-e)c''(e) \left(\frac{-U''(\bar{w})}{U'(\bar{w})^3} + \frac{U''(\underline{w})}{U'(\underline{w})^3} \right),$$

$$(B.6) \quad \frac{\partial^2 K}{\partial e^2} = c''(e) \left[\frac{2-3e}{U'(\bar{w})} - \frac{1-3e}{U'(\underline{w})} \right] + e(1-e)c'''(e) \left[\frac{1}{U'(\bar{w})} - \frac{1}{U'(\underline{w})} \right]$$

$$(B.7) \quad + e(1-e)c''(e)^2 \left[\frac{-(1-e)U''(\bar{w})}{U'(\bar{w})^3} - \frac{eU''(\underline{w})}{U'(\underline{w})^3} \right].$$

When Assumption B.1 holds, $\frac{\partial^2 K}{\partial e^2} \geq 0$.

Proof. The KKT conditions yield that (A-IR) always binds. Together with (A-IC), this uniquely determines $\underline{w}$. The constraint (A-IC) then uniquely determines $\bar{w}$. Uniqueness follows from $U' > 0$. The expressions for $\frac{\partial^2 K}{\partial e \partial u}$ and $\frac{\partial^2 K}{\partial e^2}$ arise from implicit differentiation of equations (B.3) and (B.4).

Consider $\frac{\partial^2 K}{\partial e^2}$:

$$\begin{aligned} \frac{\partial^2 K}{\partial e^2} = & \underbrace{2c''(e)\frac{e}{U'(\underline{w})}}_{(A)} + \underbrace{e \left[\frac{-c''(e) + (1-e)c'''(e)}{U'(\bar{w})} \right] + (1-e) \left[\frac{-ec'''(e) - c''(e)}{U'(\underline{w})} \right]}_{(B)} \\ & + \underbrace{\frac{2c''(e)(1-e)}{U'(\bar{w})}}_{(C)} - \underbrace{\frac{U''(\bar{w})c''(e)^2(1-e)^2e}{U'(\bar{w})^3} - \frac{U''(\underline{w})c''(e)^2(1-e)e^2}{U'(\underline{w})^3}}_{(D)} \end{aligned}$$

By $U' > 0, U'' < 0$, term (D) is positive. To show the sum of terms (A) and (B) is positive, note first that by Assumption B.1,

$$-c''(e) + (1-e)c'''(e) = c''(e)(1-e) \left(-\frac{1}{1-e} + \frac{\partial}{\partial e} [\log c''(e)] \right) \geq 0.$$

Because $U'' < 0$, a sufficient condition for (A) to be positive is:

$$\frac{2c''(e)e}{U'(\underline{w})} + e \frac{-c''(e) + (1-e)c'''(e)}{U'(\underline{w})} - (1-e) \frac{ec'''(e) + c''(e)}{U'(\underline{w})} \geq 0.$$

The above simplifies to $(2e-1)c''(e) \geq 0$, which holds for $e \geq \frac{1}{2}$. Since term (C) is positive, convexity holds for $e \geq \frac{1}{2}$.

For $e \leq \frac{1}{2}$, $U'' < 0$ implies the following relation holds:

$$\frac{2c''(e)(1-e)}{U'(\bar{w})} \geq \frac{2c''(e)e}{U'(\underline{w})}$$

Thus, by using the same process as above, the sums of terms (B) and (C) is positive. Because terms (D) and (A) are positive by $U' > 0$ and $U'' < 0$, convexity holds for $e \leq \frac{1}{2}$. $\square$

Agent Protected by Limited Liability I show that when the agent is protected by limited liability and under Assumption B.1, the principal strictly benefits from a positive value of the agent's outside option. With the restriction of limited liability, this case is similar to that where the agent's utility exhibits Constant Relative Risk Aversion (CRRA) with a parameter $\eta < 1$. In this case, the agent's utility takes the form $U(x) = \frac{x^{1-\eta}}{1-\eta}$. Because the agent's utility is only defined for positive transfers, this case can be interpreted as one

in which the intermediary implicitly faces a limited liability constraint from the agent.

The intermediary receives contract $(\underline{t}, \bar{t})$ from the principal, where $\bar{t}$ is paid upon project success, and $\underline{t}$ is paid upon project failure. The intermediary chooses $(\underline{w}, \bar{w})$ to solve

$$\begin{aligned} \max_{e, \bar{w}, \underline{w}} \quad & \{e(\bar{t} - \bar{w}) + (1 - e)(\underline{t} - \underline{w})\} \\ \text{s.t.} \quad & U(\bar{w}) - U(\underline{w}) = c'(e) \quad (\text{A-IC}) \\ & eU(\bar{w}) + (1 - e)U(\underline{w}) - c(e) \geq u \quad (\text{A-IR}) \\ & \bar{w}, \underline{w} \geq 0 \quad (\text{A-LL}) \end{aligned}$$

Note again that the convexity of c implies that the agent's FOC is a sufficient condition for (A-IC).

Proposition B.4. *Suppose Assumption B.1 holds. When the agent is protected by limited liability, there exists $u' > 0$ such that the principal strictly prefers the agent's outside option to be u' than zero, i.e., $V(u') > V(0)$.*

Proof. Upon receiving the principal's contract, the intermediary's KKT conditions yield two candidate solutions:

1. **Contract 1:** (A-IR) binds. Effort e must solve:

$$(B.8) \quad \bar{t} - \underline{t} = \frac{\partial K}{\partial e} \Big|_{(u, e)}.$$

2. **Contract 2 (Success-contingent contract):** (A-LL) binds. the intermediary optimally incentivizes the agent's effort e by offering the agent a wage $h(c'(e))$ for successful output, and zero otherwise, where $h(x) := U^{-1}(x)$. Moreover, if (A-IR) does not bind, effort e must solve

$$(B.9) \quad \bar{t} - \underline{t} = h(c'(e)) + eh'(c'(e))c''(e)$$

If (A-IR) binds, effort e must solve $ec'(e) - c(e) = u$.

Fix $u = 0$. Suppose the intermediary receives contract $(\underline{t}^0, \bar{t}^0)$ from the principal and optimally incentivizes e^0 . When $u = 0$, (A-IR) cannot bind without setting $\underline{w} < 0$; however, the agent's LL constraint precludes this. Thus, the intermediary offers Contract 2.

Define $u'_\epsilon := e^0 c'(e^0) - c(e^0) + \epsilon$, where $\epsilon > 0$ and $u'_\epsilon \leq e^0 \bar{t}^0 + (1 - e^0) \underline{t}^0 - c(e^0)$. By the ability of the intermediary to extract rents at $u = 0$, such an ϵ exists. Moreover, define $\tilde{e}$ to solve $u'_\epsilon = \tilde{e} c'(\tilde{e}) - c(\tilde{e})$. Since the RHS is strictly increasing in e , it holds that $\tilde{e} > e^0$.

Now, consider the intermediary's contract choice at $u'_\epsilon > 0$. The candidate solutions are:

1. **Contract 1:** Offer a contract s.t. the agent's IR constraint binds and offer a positive transfer upon no output. In other words, offer $h(x_L)$, where $x_L := u'_\epsilon + c(e) - ec'(e)$, for low output and $h(x_H)$, where $x_H := u'_\epsilon + c(e) + (1 - e)c'(e)$, for high output. By Lemma B.1 the intermediary's expected cost of incentivizing effort e when (A-IR) binds is $K(e, u) := eh(x_H) + (1 - e)h(x_L)$.

2. **Contract 2:** Incentivize $\tilde{e}$ with transfer $h(c'(\tilde{e}))$ for successful output and 0 otherwise.

I show that for ϵ sufficiently small, effort in the first candidate contract exceeds e^0 . First, note that the characterization of the optimal $\bar{w}, \underline{w}$ in Lemma B.1 imply that $\frac{\partial K}{\partial e}$ is continuous in u . Then, taking the limit of $\frac{\partial K}{\partial e}$ at (u'_ϵ, e^0) as ϵ approaches 0,

$$\begin{aligned} \lim_{\epsilon \rightarrow 0} \frac{\partial K}{\partial e} \Big|_{(u'_\epsilon, e^0)} &= h(c'(e^0)) + e^0(1 - e^0)c''(e^0) \left(h'(c'(e^0)) - \lim_{\epsilon \rightarrow 0} h'(\epsilon) \right) \\ &< h(c'(e^0)) + e^0(1 - e^0)c''(e^0)h'(c'(e^0)), \end{aligned}$$

where second inequality follows from $h' > 0$. Compare the limit to the following:

$$\begin{aligned} \frac{\partial}{\partial e} [eh(c'(e))] \Big|_{e=e^0} &= h(c'(e^0)) + e^0 c''(e^0) h'(c'(e^0)) \\ &> \lim_{\epsilon \rightarrow 0} \frac{\partial K}{\partial e} \Big|_{(u'_\epsilon, e^0)}. \end{aligned}$$

By the continuity of both derivatives, there exists $\epsilon' > 0$ such that

$$\frac{\partial}{\partial e} [eh(c'(e))] \Big|_{e^0} > \frac{\partial K}{\partial e} \Big|_{(u'_{\epsilon'}, e^0)}.$$

By the necessity of the FOC, established in equation (B.9),

$$(B.10) \quad \bar{t} - \underline{t} = \frac{\partial}{\partial e} [eh(c'(e))] \Big|_{e=e^0} > \frac{\partial K}{\partial e} \Big|_{(u'_{\epsilon'}, e^0)}.$$

By the FOC given in equation (B.9), the convexity of K determined by Lemma B.1, and the inequality in equation (B.10), then if Contract 1 is optimal, then it must incentivize effort $\hat{e} > e^0$. Additionally, note that since $\tilde{e} > e^0$, Contract 2 incentivizes a strictly higher effort than under $u = 0$.

In both candidate contracts, effort is strictly higher at u'_ϵ than at $u = 0$. Moreover, because the intermediary achieves a positive payoff at $u = 0$, there exists a feasible contract at u'_ϵ that achieves the intermediary's nonnegative utility. It follows that $V(u'_\epsilon) > V(0)$.

□

Agent Not Protected by Limited Liability I show that when the agent is not protected by limited liability, the principal's ability to benefit from higher values of the agent's outside option depends on the expected cost of incentivizing effort.

Assume $U: \mathbb{R} \rightarrow \mathbb{R}$. The intermediary receives contract $(\underline{t}, \bar{t})$ from the principal, where $\bar{t}$ is paid upon project success, and $\underline{t}$ is paid upon project failure. Because the intermediary does not face a limited liability constraint and $K(e, u)$ is the lowest expected cost of incentivizing effort e given outside option u , the intermediary solves

$$\max_e \{e\bar{t} + (1-e)\underline{t} - K(e, u)\}.$$

Proposition B.5. *Let e^* be the unique solution to the intermediary's problem given u . If $\frac{\partial^2 K}{\partial e \partial u} |_{(e^*, u)} < 0$, then there exists $u' > u$ such that $V(u') > V(u)$. If $\frac{\partial^2 K}{\partial e \partial u} \geq 0$ for all (e, u) , then the principal weakly prefers the agent's outside option to be zero.*

Proof. Given the principal's contract $(\bar{t}, \underline{t})$, let e_I^* solve the intermediary's problem, which implies it necessarily solves his FOC:

$$\bar{t} - \underline{t} = \left. \frac{\partial K}{\partial e} \right|_{e=e_I^*}.$$

Uniqueness follows from the convexity of K in e , shown by Lemma B.1. Taking the implicit derivative with respect to u yields

$$\frac{\partial e_I^*}{\partial u} = - \frac{\frac{\partial^2 K}{\partial e \partial u} |_{(e_I^*, u)}}{\frac{\partial^2 K}{\partial e^2} |_{(e_I^*, u)}}.$$

By Lemma B.1, $\frac{\partial^2 K}{\partial e^2} |_{(e_I^*, u)} > 0$.

If $\frac{\partial^2 K}{\partial e \partial u} |_{(e_I^*, u)} < 0$ then $\frac{\partial e_I^*}{\partial u} > 0$. It follows from the continuity of e_I^* that $\exists u' > u$ such that at u' , the solution to the intermediary's problem e_I' satisfies $e_I' > e_I^*$. Thus, the principal can incentivize strictly higher effort at the same transfer, so $V(u') > V(u)$.

If $\frac{\partial^2 K}{\partial e \partial u} \geq 0$ for all (e, u) , then given the principal's contract, the intermediary incentivizes weakly lower effort under u' than under $u = 0$ for all values of the principal's transfer t . Thus, the principal incentivizes weakly lower effort given all transfers, and it follows that $V(0) \geq V(u')$ for all $u' \geq 0$. $\square$

Examples. Consider the following utility functions:

1. **Constant Absolute Risk Aversion (CARA):** $U(x) = \frac{1-e^{-ax}}{a}$ for $a \neq 0$, $U(x) = c$ for $a = 0$.

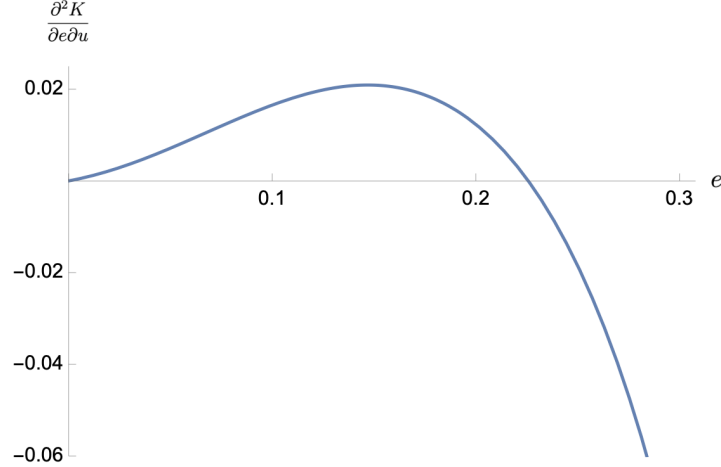


Figure 5: The cross partial of the expected cost of incentivizing effort $\frac{\partial^2 K}{\partial e \partial u}$ evaluated at agent outside option $u = 0.2$.

2. Constant Relative Risk Aversion (CRRA): $U(x) = \frac{x^{1-\eta}}{1-\eta}$ for $\eta \geq 0, \eta \neq 1$ and $U(x) = \log x$ for $\eta = 1$.

Lemma B.2 shows that for CARA utility, $\frac{\partial^2 K}{\partial e \partial u} \geq 0$ for all (e, u) , so the principal cannot benefit from higher values of the agent's outside option. Following the result, I show an example that exhibits how CRRA utility functions can exhibit $\frac{\partial^2 K}{\partial e \partial u} |_{(e,u)} < 0$.

Lemma B.2. *Given CARA utility, $\frac{\partial^2 K}{\partial e \partial u} \geq 0$ for all (e, u) . Moreover, the principal weakly prefers the agent's outside option to be zero, i.e. $V(0) \geq V(u) \forall u \geq 0$.*

Proof. Given the functional form of CARA,

$$\frac{\partial}{\partial x} \left[\frac{U''(x)}{U'(x)^3} \right] = -2a^2 e^{2ax} \leq 0.$$

Because the KKT conditions yield $\bar{w} > \underline{w}$, the inequality above implies $\frac{-U''(\bar{w})}{U'(\bar{w})^3} + \frac{U''(\underline{w})}{U'(\underline{w})^3} \geq 0$. Moreover, because $U'(x)$ is strictly decreasing in x , $\frac{1}{U'(\bar{w})} - \frac{1}{U'(\underline{w})} \geq 0$. It follows from equation (B.5) that $\frac{\partial^2 K}{\partial e \partial u} \geq 0$ for all (e, u) . The result follows from Proposition B.5. $\square$

Example: CRRA Utility. Consider CRRA utility and let $c(x) = x^3, \eta = 0.1$. Suppose $u = 0.2$. Note that due to (A-IR), the results of Lemma B.1 carry for $e \in [0, 10^{-1/3}]$. Using equation (B.5), the parameters of the problem yield $\frac{\partial^2 K}{\partial e \partial u} |_{u=0.2} < 0$ for $e \in [0.23, 0.46]$.

C. Supplementary Material for Continuous Output Proofs

The following proofs supplement those for the case of continuous output. I show that at $u = 0$, the principal optimally incentivizes strictly positive effort in both the intermediated and second-best settings.

Lemma C.1. *Consider the case of continuous output. When $u = 0$, the principal incentivizes positive effort in second-best.*

Proof. When $u = 0$, both the principal and the intermediary are unconstrained by (A-IR). The principal's cost of incentivizing effort e is $K(e)$. By Lemma B.1, the solution to this program for $e = 0$ yields a l.o.d. contract with cutoff $z_P = 1$ and a multiplier $\lambda_{A-IC} \geq 0$. By the Envelope Theorem,

$$\frac{\partial K}{\partial e} = \int_{z_P}^{\bar{y}} y f_e(y | e) dy + \lambda_{A-IC} \left(\int_{z_P}^{\bar{y}} y f_{ee}(y | e) dy - c''(e) \right).$$

Plugging in $z_P = 1$, $K'(0) \leq 0$.

Now, consider the principal's expected value in second-best as a function of effort induced: $v(e) := \int_0^{\bar{y}} y f(y | e) dy - K(e)$. By MLRP and $K'(0) \leq 0$, it follows that $v'(0) > 0$. Thus, the principal can do strictly better by incentivizing positive effort. $\square$

Lemma C.2. *Consider the case of continuous output. When $u = 0$, and under Assumptions 1-2, the principal incentivizes positive effort when contracting is intermediated.*

Proof. When $u = 0$, both the principal and the intermediary are unconstrained by (A-IR). The KKT conditions yield that given t , the solution to the intermediary's problem is a l.o.d. contract with cutoff z_I and effort e such that $(z_I, e) \in \text{Slack}(t)$. Define the following maximization problem, which is a relaxation of the principal's cost of incentivizing effort e :

$$\begin{aligned} K_I(e) &= \min_{t \in \mathcal{T}, z_I} \int_0^{\bar{y}} t(y) f(y | e) dy \\ \text{s.t.} \quad & (z_I, e) \in \text{Slack}(t) \quad (\text{A-IC}). \end{aligned}$$

By Lemma B.7 and B.10, the principal's optimal contract is a l.o.d. contract w^{sLOD} with cutoff z_P ; moreover, $f_e(z_P | e) > 0$. By Lemma B.4, $\text{Slack}(w^{sLOD})$ is single-valued. Thus, the solution to this relaxed program is the principal's minimal cost of incentivizing effort e . The solution to this program for $e = 0$ yields $t = 0$, $z_I = 1$. By the Envelope Theorem, $K'(0) \leq 0$.

The principal's expected value as a function of effort induced is $v(e) = \max_{e'} \left\{ \int_0^{\bar{y}} y f(y \mid e') dy - K_I(e') \right\}$. By MLRP and $K'(0) \leq 0$, it follows that $v'(0) > 0$. Thus, the principal can do strictly better by incentivizing positive effort. $\square$